# Simulating Filtration Processes using a Network Model

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**Abstract**

Granular filter design rules have generally been empirical; a need arises to understand filtration processes at particulate level. Although particle-based methods such as the coupled Discrete Element Method and Computational Fluid Dynamics (DEM-CFD) offer perhaps the most complete account of filtration processes, the increased computational cost limits the number of particles that can be simulated simultaneously. This research, therefore, proposes a network model that allows computations to be performed at pore-scale. The model incorporates two novel features, specifically realistic fluid flows and accurate void space structure, that are made possible with the development of the DEM and the Delaunay Triangulation Method. By comparing filtration processes simulations based on networks that match the actual void space to varying degree, this research demonstrates the significance of modelling realistic void space structure. The proposed network model is also able to provide insights into the fundamental mechanics of filtration processes, and to assess the accuracy of established design criteria.

## List of Abbreviations

| | |
|---|---|
| CFD | Computational fluid dynamics |
| CIRIA | Construction Industry Research and Information Association |
| CSD | Constriction size distribution |
| DEM | Discrete Element Method |
| DTM | Delaunay Triangulation Method |
| FEMA | Federal Emergency Management Agency |
| ICM | Inscribed Circle Method |
| USBR | United States Bureau of Reclamation |
| PSD | Particle size distribution |

## List of Symbols

| | |
|---|---|
| $A$ | Total cross-sectional area of edge |
| $\alpha$ | Shape factor |
| $\beta$ | Ratio of reduced to total cross-sectional area of edge |
| $C_u$ | Coefficient of uniformity |
| $d^n$ | Diameter of base particle normalised by $D_{0F}$ |
| $d^n_{largest}$ | Diameter of the largest base particle eroded normalised by $D_{0F}$ |
| $d_{xB}$ | Particle diameter for which $x$ percent of base material is finer by mass |
| $D_{xF}$ | Particle diameter for which $x$ percent of filter material is finer by mass |
| $D^*_{xc}$ | Constriction size for which $x$ percent is smaller by number |
| $e$ | Void ratio |
| $g$ | Local conductance of edge |
| $k$ | Permeability |
| $L$ | Length of edge |
| $\mu$ | Coefficient of friction adopted in Discrete Element Method (DEM) or fluid viscosity |
| $p$ | Piezometric pressure |
| $p_{eroded}$ | Percentage of base particles that are eroded by volume |
| $\Phi$ | Potential or gravitational field |
| $\phi$ | Packing density |
| $Q$ | Total fluid flow |
| $q$ | Fluid flow per unit area |
| $R_d$ | Relative density |
| $R^h$ | Hydraulic radius of edge |
| $r$ | Radius of edge |
| $\rho$ | Fluid density |
| $S$ | Surface area of fluid-sold interface |
| $s$ | Fluid source |
| $S_v$ | Specific internal surface area to volume ratio from PSD |
| $T_{intro}$ | Number of time steps for which base particles are introduced |
| $u$ | Fluid velocity |
| $V$ | Volume |

## 1. Introduction

### *1.1. Background*

In hydraulic earth structures, granular filters are employed to prevent erosion while enabling water to flow. Typically made of coarsely-grained soils such as cohesionless sand or gravel, a granular filter is placed downstream of and adjacent to the base soil to be retained (see Figure 1(a)). A granular filter is also referred to as a filter-drain because it is intended not only to prevent the loss of material, but also to maintain seepage pressure at an acceptable level (FEMA 2011). This research focuses on the filtration function, that is, the effectiveness of a filter in retaining base particles.

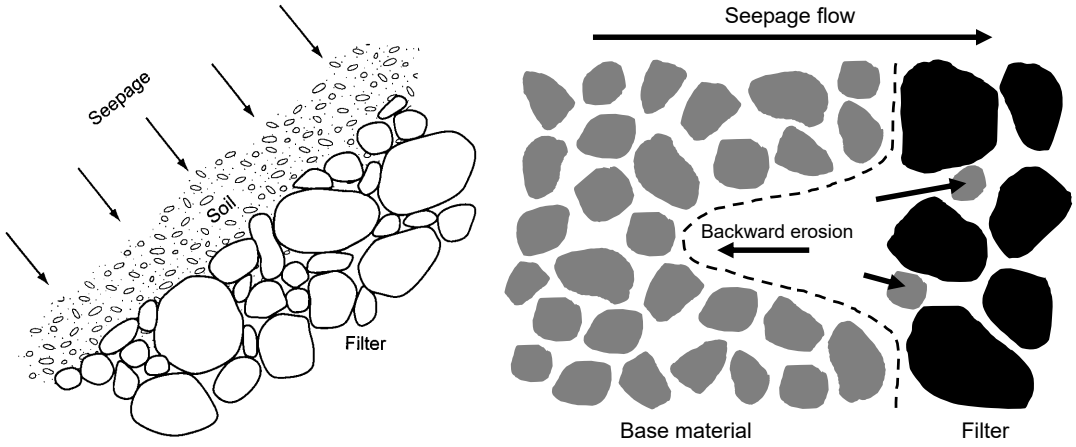



Figure 1: Schematics showing (a) how an effective filter prevents movement of base particles (FEMA 2011), and (b) the mechanism of backward erosion (adapted from CIRIA et al. (2013)).

Assuming that the base particles are internally stable, the use of ineffective filters would lead to backward erosion, also known as piping (see Figure 1(b)). The process of backward erosion is progressive, with a few of the base particles at the interface with filter being eroded initially. The defect in the base material then leads to greater seepage, resulting in further erosion and the formation of a cavity or pipe. If unnoticed, this could lead to deterioration of the earth structures and ultimately structural failure in severe cases.

The most common hydraulic earth structures of interest in civil engineering are embankment dams and flood embankments. The typical components, including the filter, of an embankment dam are shown in Figure 2. A statistical analysis performed by Foster et al. (2000) showed that piping is one of the main causes of failure in embankment dams constructed up to year 1986, accounting for approximately 50 percent of such occurrences. Due to the large amount of water retained, the failure of such infrastructures can be catastrophic especially if the area downstream is populated. This is unfortunately observed in Gouhou, China, where the failure of the 71 m high concrete-faced rockfill dam resulted in 300 deaths (Zhang & Chen 2006). In addition, the suspension of hydroelectric power stations in dams may lead to disruption as they are major contributors to power supply, e.g. WAC Bennett dam in Canada (see Figure 3).

Granular filters are also employed in dewatering wells (see Figure 4), which are used to lower the surrounding water table through constant pumping of water. This leads to continuous seepage into the well. If the filters are not appropriately designed, the loss of material would cause significant settlement around the well and thus damage to adjacent

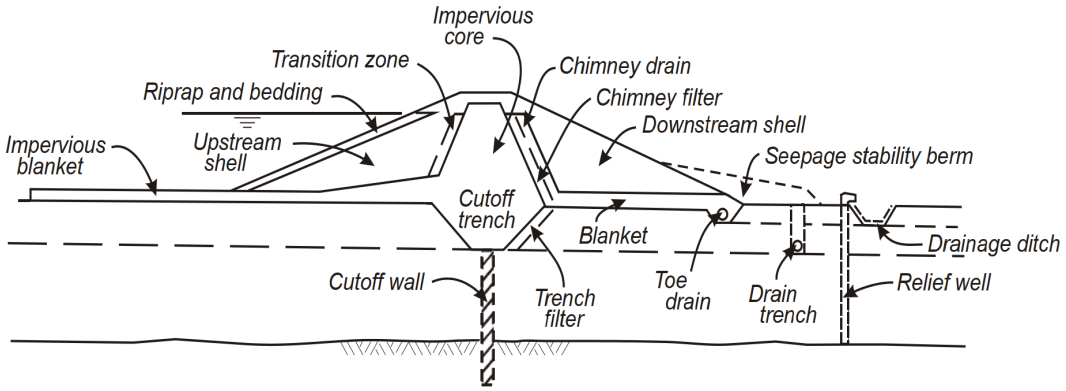



Figure 2: Typical components of embankment dams (FEMA 2011).



Figure 3: Sinkhole observed at WAC Bennett Dam in Canada (Muir Wood 2007).

infrastructures and services. Furthermore, large amount of particles that are suspended in the water would damage the pumps.

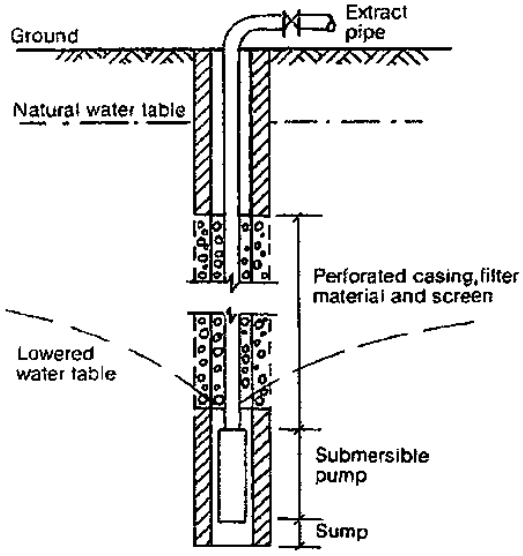



Figure 4: Granular filter in a dewatering well (Engineers Daily n.d.).

The design frameworks of filters adopted in guidelines have generally been empirical (see USBR (1987), FEMA (2011) and CIRIA et al. (2013)). The most notable being Terzaghi's filter rule (Terzaghi & Peck 1948), which characterises the effectiveness of filters by comparing representative particle sizes of the base soil and the filter. Although this method is simple to apply, it does not account for important factors of the filters

such as the coefficient of uniformity $C_u$ and the void ratio $e$. Furthermore, empirical design rules are strictly only applicable to the tested soils, hence could be inappropriate for special cases. At the other end of the complexity spectrum, particle-based methods, such as the coupled discrete element method and computational fluid dynamics, offer perhaps the most complete account of the filtration processes. These methods simulate individual soil particles and the interaction forces between each particle, as well as the hydraulic forces acting on the particles. Obtaining the hydraulic forces involves discretising the fluid domain into smaller elements, either on a coarse-grained level (e.g. Huang et al. (2014)) or on a fully-resolved level, and subsequently calculating the pressure and velocity fields. This allows the effects of additional factors such as the hydraulic conditions and the confining stresses within the filter on the extent of erosion to be evaluated. However, the increased computational cost limits the number of particles that can be considered simultaneously, especially when simulating long time-scale events such as filtration processes. Efforts have been made to reduce this cost by simplifying the calculations of fluid forces (see Chareyre et al. (2012) and Sufian et al. (2019)).

Simulation of filtration processes using a network model firstly entails representing the void space of the filter as a network, i.e. a set of nodes connected by edges. This then allows the calculation of flows and the simulation of particle movements within the filter to be performed at pore-scale, reducing the computational demand relative to particle-based methods. Thus, network models provide an attractive computational alternative. This approach also involves assigning representative geometry to the network, and therefore can be used to better understand the effects of varying filter $C_u$ and $e$ on the filter effectiveness. In addition, the simulation of individual particles allows time-dependent behaviours such as the development of self-filtration, as well as the changes in void sizes and flow rates to be captured. It is important to note that self-filtration is the process in which retained coarser base particles are able to prevent the erosion of other finer base particles. Unlike experimental means, the fundamental behaviours of individual particles within a filter can be easily observed and tracked for analysis, the results of which should provide insights into how effective filters work in comparison with prior studies (e.g. Shire & O'Sullivan (2017)).

The application of network models to study filtration processes is not new (e.g. Rege & Fogler (1987)). The models used to date invoked idealising assumptions that often oversimplify the structure of the void space. The development of the Discrete Element Method (DEM) and the Delaunay Triangulation Method (DTM) provides a means by which accurate void space structure can be quantified. This process is suitable for the construction of network models since it generates not only the the positions of the pores and constrictions, but also geometric properties such as the volume and the surface area of each node, as well as the cross-sectional area and the length of each constriction.

### *1.2. Aim and Objectives*

The aim of this research is to investigate filtration processes at pore-scale using network model simulations. To achieve this, the specific objectives are defined as follows:

1. To develop a network model that incorporates realistic features such as consideration of fluid flows and accurate description of void space structure based on the DTM-derived data. The data is provided by Dr. Adnan Sufian based on virtual filter samples created using DEM simulations by Shire (2014).

2. To simulate filtration processes using networks that match the actual void space to varying degree, and to evaluate the significance of modelling realistic void space

structure by comparing the simulated results between the different networks and against those obtained experimentally or numerically.

3. To perform quantitative analysis of the fundamental mechanics of filtration processes by simulating the most realistic network, and to examine how filter and base characteristics affect these processes.

4. To assess the accuracy of existing granular filter design criteria proposed by Terzaghi & Peck (1948), Foster & Fell (2001) and Indraratna et al. (2007).

### *1.3. Outline*

The remaining of the paper is divided into four sections:

Section 2 - Literature review of established practices used in the design of granular filters, as well as of existing researches on the application of network models in simulating filtration processes.

Section 3 - Description of the network model to be adopted in the research, and the implementation of the model in MATLAB.

Section 4 - Application of the network model and discussions of results.

Section 5 - Presentation of conclusions and potential direction for future research.

## 2. Literature Review

### *2.1. Granular Filter Design*

Within a granular filter, the separation of finer particles from the fluid can occur through different mechanisms. These include 'straining, frictional and/or cohesive trapping, interception, flocculation, sedimentation, adsorption, diffusion, as well as hydrodynamics and inertia' (Roozbahani et al. 2014). Physical straining, also referred to as mechanical trapping, of the particles is the focus of current research. This process is generally considered to occur when the constrictions within the void space are smaller than the particles carried along by fluid flow. Within the context of particle straining, Kenney et al. (1985) listed three independent factors affecting the effectiveness of granular filters: the geometry of the void space of the filter material, the sizes of the base particles and the hydraulic conditions such as the velocity and the direction of the seepage.

The most widely used filter rule was proposed by Terzaghi, where a filter is considered effective if the following inequality is met (Terzaghi & Peck 1948):

$$\frac{D_{15F}}{d_{85B}} \leq 4 \tag{1}$$

where $x$ percent of filter particles by mass is finer than $D_{xF}$, while $y$ percent of base particles by mass is finer than $d_{yB}$. The rule is derived based on the assumption that $D_{15F}/5$ is representative of the controlling constriction size of the filter layer (Fannin 2008). The concept of a controlling constriction diameter was explained by Kenney et al. (1985), who proposed that the controlling constriction size is equal to the largest particle diameter that is eroded, and that this constriction size is primarily determined by the smaller particle diameters of the filter (i.e. within the range of $D_{5F}$ to $D_{15F}$) instead of the shape of the particle size distribution (PSD) curve. Terzaghi's rule was further advocated by Soares (1980), who performed experiments in which mechanical sieves were used as filters. These experiments show that sieves with opening diameter smaller than $d_{80B}$ have minimal erosion, whereas sieves with opening diameter larger than $d_{90B}$ have substantial particle loss. The use of relatively large base particle diameter as the characteristic value demonstrates the ability of self-filtration, i.e. the clogging of filter by larger base particles results in the retention of the smaller base particles. Although other characteristic ratios such as $D_{50F}/d_{50B}$ or $D_{15F}/d_{15B}$ have been suggested (e.g. Karpoff (1955)), Honjo & Veneziano (1989) argued that the use of these ratios are not justified as these ratios do not correlate with the filter effectiveness. Design rules proposed by Sherard & Dunnigan (1989) are also widely accepted. They suggested varying the effectiveness criterion for soils that consist of different percentages of fines.

Foster & Fell (2001) proposed a framework for assessing existing dams filters. They characterised the performance of a filter as (1) seal with no erosion, (2) seal with some erosion or (3) partial or no seal with large erosion. For base particles with fines content less than 15 percent, the boundaries between (1) and (2), as well as between (2) and (3) are given in Equation 2 and 3 respectively:

$$\frac{D_{15F}}{d_{85B}} \leq 7 \tag{2}$$

$$\frac{D_{15F}}{d_{95B}} \leq 9 \tag{3}$$

The use of representative filter particle sizes in Equations 1 to 3 implies that the

geometry of the void space can only be accounted for indirectly. Therefore, constriction based criteria were also suggested such that the effects of the filter $C_u$ and $e$ can be implicitly considered. An example is Indraratna et al. (2007) as shown below:

$$\frac{D^*_{35c}}{d_{85B(SA)}} \leq 1 \tag{4}$$

where $x$ percent of constrictions by number is smaller than $D^*_{xc}$ and $d_{85B(SA)}$ is evaluated from the PSD scaled by surface area. Indraratna et al. (2007) adopted $d_{85B(SA)}$, which is smaller than the conventional $d_{85B}$, since Honjo & Veneziano (1989) demonstrated that mechanical sieves were effective only if at least 15 percent of base particles were retained. This study will adopt a network model to assess the accuracy of the criteria set out in Equations 1 to 4.

### *2.2. Network Model*

A network model in its simplest form is a set of nodes interconnected with edges. Newman (2010) described network models as 'simplified representations that reduce systems to abstract structures capturing only the basics of connection patterns and little else', and stated that additional properties can be assigned to the nodes and edges to incorporate more details. Although information is lost in the process, the advantages are simplified calculations and thus significantly reduced computational demand (Newman 2010).

In this research, network models are used to represent the void spaces of granular filters to study the transport and filtration of the base particles. As shown in Figure 5, porous media consist of pores that are linked by narrow constrictions; these can be idealised as nodes and edges respectively. Network models are suitable as they simply describe the interconnectivity of the void space of a porous medium (Rege & Fogler 1987). Network models implicitly assume that the filter particles are stable and fixed in space, thus the overall matrix integrity is not lost and the total volume of the filter is constant throughout the filtration process.

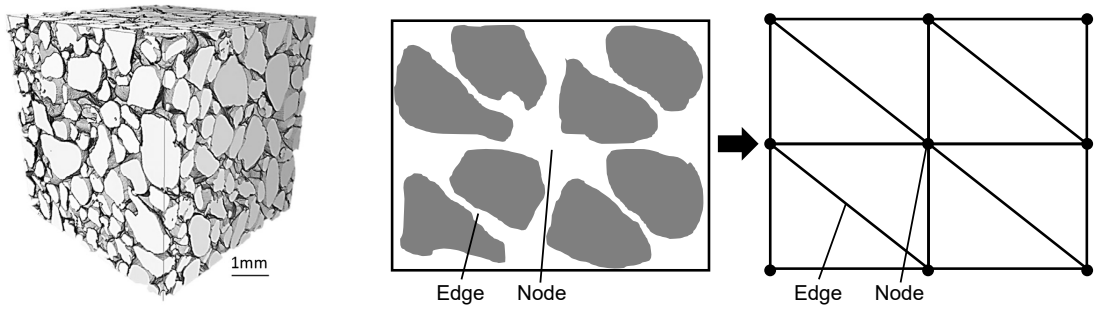



Figure 5: Representation of void space as a network. (a) Micro-CT image of Leighton Buzzard sand (Taylor et al. 2015). (b) Similarity between the net porous media and network model (adapted from Rege & Fogler (1987)).

It is worth introducing a framework to understand the components of the void space structure. With reference to Figure 6, the structure consists of topology and geometry. The network topology is the way nodes and edges are arranged. The arrangement of edges affects the network connectivity, which is defined as the number of neighbouring nodes that are connected to each node. Connectivity is also referred to as coordination number. The network geometry on the other hand can be fully described by its distributions and

spatial correlation. Distribution refers to the probabilistic distribution of a particular geometric property in the network, while spatial correlation refers to the relationships that exist between the geometric properties of neighbouring parts.

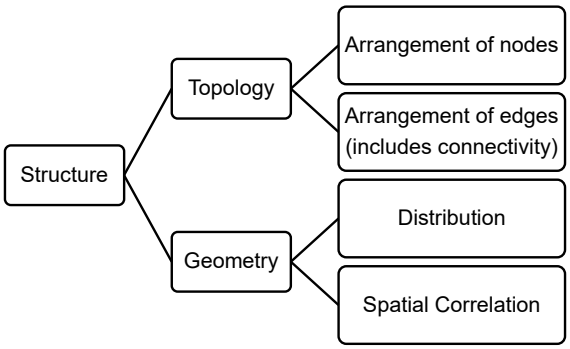



Figure 6: Components of the structure of void space.

Networks in which the nodes and edges are regularly arranged are called lattices. In such networks, all nodes except those at the boundaries typically have the same connectivity. In reality, actual porous media have networks that are highly irregular such that nodes are randomly positioned and have varying degrees of connectivity.

According to Rege & Fogler (1987), the use of a network model to represent porous media was first proposed by Fatt in 1956. Since then, it has been applied in many fields such as chemical processes, petroleum industry, water and wastewater treatment, as well as geotechnical engineering to examine the transport and filtration of particles. Examples relevant to this research are listed in Table 1.

It is critical to choose a network that best represents the structure of the void space. However, it is observed that many of the network models made idealising assumptions that often oversimplify the structure of the void space; the assumptions include:

1. The use of regular networks or lattices, e.g. square or cubic, such that all nodes have the same connectivity except those at the boundaries.

2. The geometry of the void space such as edge sizes and lengths are distributed randomly and independently in the network.

In their analysis of two-phase flow in sandstones at pore-scale, Valvatne & Blunt (2004) discovered that the use of regular networks is unsatisfactory as the results vary from those obtained with more realistic networks. They suggested that accurate distribution of void connectivity is critical to predict realistic flow properties in porous media. However, there are contradicting conclusions on the significance of accurate network connectivity on the predicted filtration behaviour. Shire & O'Sullivan (2017) justified the use of regular cubic network based on the experimental work of Witt, who discovered that there is an average of 5.7 constrictions per pore by studying resin-impregnated gravel samples. However, Shire & O'Sullivan (2017) concluded that the results are sensitive to the initial network connectivity based on a sensitivity analysis; the analysis was performed by repeating simulations on a network in which constrictions are added or removed while maintaining the same average connectivity of 6. On the other hand, Ding et al. (2018) decided to use a square network since in their previous studies it was discovered that there are insignificant differences in results when 2D and 3D networks were used.

Table 1: Network models that were adopted in previous researches.

| Research | Scope | Network Model |
|---|---|---|
| Silveira et al. (1975) | Proposed a method to estimate the infiltration depth based on the physics of mechanical trapping, i.e. a base particle is retained if it is larger than the constriction size. | Probabilistic network model |
| Rege & Fogler (1987) | Simulated straining-dominated particle entrapment to evaluate the minimum degree of prefiltration of the injection water required such that damage to formations can be prevented. | Regular 2D triangular network |
| Locke et al. (2001) | Studied the movement of particles through granular filters and the resulting time-dependent changes based on a transport model developed by Indraratna & Vafai (1997). | Regular 3D cubic network |
| Moraci et al. (2011) | Evaluated internal stability of soils. | Alternating layers of particles and constrictions |
| Shire & O'Sullivan (2017) | Assessed the effectiveness of base-filter combinations through the simulation of the movement of millions of base particles. | Regular 3D cubic network |
| Ding et al. (2018) | Examined the effects of several factors, including particle and constriction size distributions (PSD and CSD), on straining-dominated deep bed filtration. | Regular 2D square network |

As for the geometry of the void space, Bryant et al. (1993) showed that spatial correlation between geometric properties of neighbouring components exists in random monodisperse sphere packing (further discussed in Section 3.1.2). Using a network constructed based on a real random sphere packing, they were able to accurately predict the permeabilities of other packings. When they redistributed the effective radii and lengths of the edges, it was found that the permeability of the resulting network increased by 78%. The percolation threshold is the minimum number of edges in the network beyond which paths that span the network will form. Mason & Mellor (1991) carried out a similar procedure on simulations of mercury intrusion; when the edge sizes in the network are redistributed randomly, the percolation threshold decreases from 51% to 39%. It is interesting to note that diamond lattices with a connectivity of 4 has the same percolation threshold of 39%.

This study will address the simplifying assumptions applied in generating network models and determine how the accuracy of the description of the void space structure affects the prediction of filtration characteristics.

### *2.3. Flow of Particles*

Directed networks, i.e. networks in which a direction is associated to each edge, are usually adopted in the simulation of particle movements as it is sensible to assume that the base particles must move in the direction of seepage flow. In their cubic network, Locke et al. (2001) assumed that particles can move only sideways and in the direction of seepage, i.e. each internal node has one entrance and five exits. Shire & O'Sullivan (2017) on the other hand assumed that particles can only move in the positive directions of the Cartesian space, i.e. each internal node has three entrances and three exits. Rege

& Fogler (1987) proposed a more realistic approach - the fluid flow through each edge was calculated, thus establishing the directions assigned to the edges. Rege & Fogler (1987) calculated fluid flows based on Hagen-Poiseuille equation and the principle of conservation of mass. A similar approach to fluid flow calculation was adopted in this study (as detailed in Section 3.3).

The calculation of fluid flow allows the concept of 'flow-biased probabilities', suggested by Rege & Fogler (1987), to be applied. The concept is used to determine the paths taken by particles that are located in nodes with multiple exits. With reference to Figure 7, the probability of exiting through a path is proportional to the flow rate in the path because the concentration of particles in the fluid should remain constant. This method not only accounts for the stochastic nature of particle movements, but also the effects of flow rate. Shire & O'Sullivan (2017) adopted an area-biased approach, which was based on the assumption that the flow in each edge is proportional to its area.

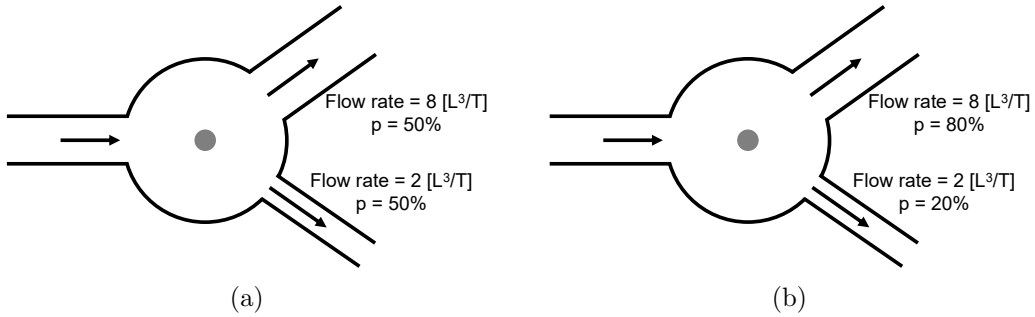
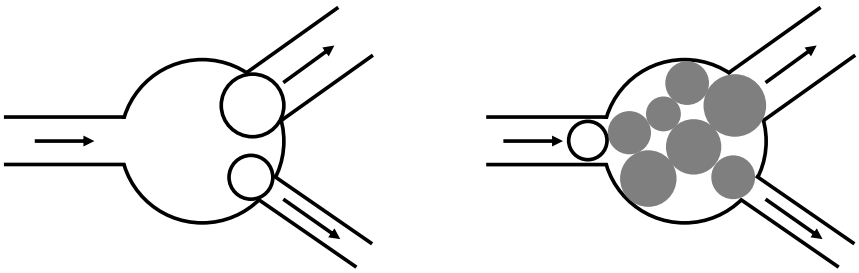



Figure 7: Comparison of (a) random walk and (b) flow-biased walk ($p$ represents probability).

In the network models used by Rege & Fogler (1987), Shire & O'Sullivan (2017) and Ding et al. (2018), multiple base particles are introduced into the system simultaneously so that time-dependent behaviours such as changes in void sizes and flow rates can be considered. These network models also simulate the process of particle straining when a particle meets a smaller constriction. When this occurs, the exit is then considered blocked, preventing other particles from passing. The process will be referred to as 'edge clogging' (see Figure 8(a)). The flow rate in each edge is then updated accordingly.



(a) Edge clogging. (b) Node clogging.

Figure 8: Mechanisms of particle filtration.

In addition, the network model by Shire & O'Sullivan (2017) accounts for the particles that are retained when all exits associated to a node are blocked or clogged. However, since the node volumes are not assigned, unrealistically large number of particles can be retained in the same node. Ding et al. (2018) distributed appropriate node volumes randomly throughout the system, thus implicitly accounted for another process which will be referred to as 'node clogging', as shown in Figure 8(b).

*2.4. Network Topology and Geometry*

Most of the previous network models are concerned only with the constriction sizes in granular filters as it is generally accepted that the effectiveness of a filter is governed by the filter CSD, which is indirectly related to the PSD. Note that constriction size is defined as the diameter of the largest circle that can fit through each constriction. Different methods have been developed to estimate the filter CSD. Among these are analytical methods such as the inscribed circle method (Silveira et al. (1975), Locke et al. (2001) and Indraratna et al. (2007)) and the Delauney Triangulation Method (DTM), with vertices defined either at the centroids of particles (Reboul et al. 2010) or at the contact points between particles (O'Sullivan et al. 2015). Image-based methods have also been adopted; for example, the maximal ball method (Dong 2008), the medial axis method (Lindquist et al. 2000) and the watershed-based method (Taylor et al. 2015).

This research considered virtual filter samples created using Discrete Element Method (DEM) simulations by Shire (2014) and also considered by Shire & O'Sullivan (2017); these DEM simulations were performed using an open-source code LAMMPS (Plimpton 1995), which allowed isotropic compression with periodic boundaries. This prevented rigid boundaries from affecting the arrangement of particles. By adjusting the coefficient of inter-particle friction $\mu$, samples of varying $e$ were produced. The samples generated with the values of $\mu$ of 0.0, 0.1, and 0.3 are referred to as 'dense', 'medium' and 'loose' respectively.

The DTM was then applied as documented in Sufian et al. (2015). Two types of DTMs, namely the weighted DTM (wDTM) by Reboul (2008) and the modified DTM (mDTM) by Al-Raoush et al. (2003), were used here. Both methods involve the tessellation of the filter space into most-regular tetrahedral void cells, of which the vertices are the particle centroids. Each vertex, i.e. the centroid of each particle, is weighted with the particle radius to reduce the risk of forming tetrahedra that are very flat or forming invalid constrictions compared to the classical Delaunay Triangulation Method (Shire 2014). The void cells are defined as the nodes, while the constrictions formed by three particles at the faces are defined as the edges. The difference between the methods is that the mDTM merges the individual void cells of which the largest inscribed spheres overlap by any amount as shown in Figure 9 (Al-Raoush et al. 2003). The properties that can be extracted are listed in Table 2.

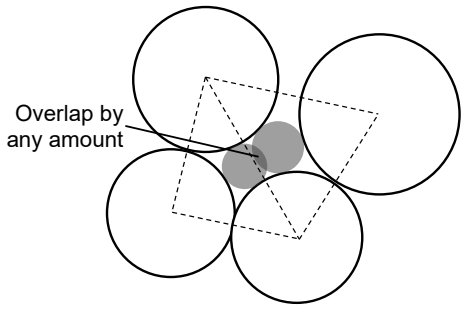



Figure 9: 2D representation of the merging criterion (adapted from Sufian et al. (2015)).

The modelling of filter particles as perfect spheres is acceptable because Taylor (2016) demonstrated that both the CSDs of packings of glass beads and of real sands are within the same range using watershed-based method as shown in Figure 10. However, Taylor (2016) noted that the CSDs of glass beads samples are more uniform than those of real sands samples due to less smaller and also less larger constrictions, resulting in

Table 2: Properties that can be extracted from DTM.

| Component | Properties |
|---|---|
| Node | Coordinates of centroid, volume, surface area |
| Edge | Total cross-sectional area, length, particles forming the constriction |

steep central portion. Taylor (2016) suggested that this is caused by both the ability of irregularly-shaped sand particles to interlock such that it is more likely to encounter smaller constrictions, and the increased likelihood of large pores with larger constrictions forming.

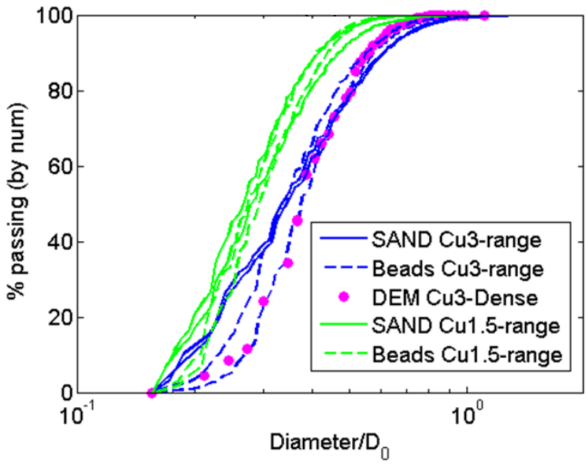



Figure 10: Comparison of CSDs of packings of glass beads and of real sands (Taylor 2016).

## 3. Model Formulation

### *3.1. Void Space Structure*

This research aims to construct more realistic networks based on actual void space structure. This was achieved by considering the virtual filter samples created using DEM simulations by Shire (2014). The samples with varying $C_u$ and $e$ considered are detailed in Figure 11 and Table 3. To explore the influence of sample length and width, the overall size of each sample was artificially increased by replicating and stacking (discussed further in Section 3.2.2). This was possible due to the use of periodic boundaries. For each filter sample, sizes of [1,1,4] and [2,2,4] were considered, where the three elements enclosed by the square brackets represent the number of filter samples stacked in x-, y- and z-directions respectively. However, the size of [2,2,4] was not possible for filter samples with $C_u = 3.0$ due to memory limitation. Note that the Dirichlet boundary condition, i.e. the pressure gradient, was imposed such that the seepage flows in the positive z-direction.

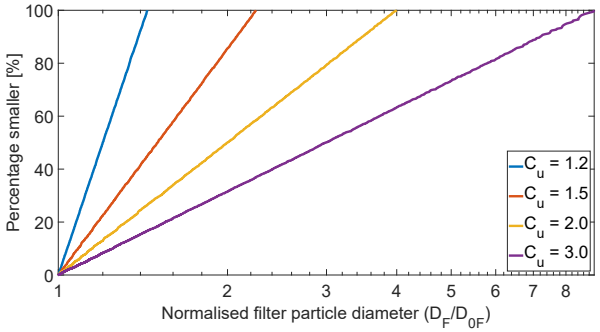



Figure 11: PSDs of virtual filter samples created by Shire (2014).

The DTM was then applied to filter particles within a smaller boundary that has a size of 80% of the original to improve the stability of the algorithm. The use of DTM allowed not only the identification of the pores and of the constrictions linking the pores, but also the quantification of the void space geometry as documented in Sufian et al. (2015). As discussed in Section 2.4, two types of DTMs were applied; the wDTM was applied to all filter samples, whereas the mDTM was applied only to dense filters of size [1,1,1]. The application of DTM and the corresponding output (listed in Table 2) were performed and provided by Dr. Adnan Sufian.

#### *3.1.1. Topology*

In Table 4, properties related to the void space topology of the virtual filter samples considered are tabulated. These properties include the total number of pores represented as nodes, the total number of constrictions represented as edges, and the average connectivity or coordination number of each node. The connectivity is slightly smaller than 4 for nodes identified with the wDTM as expected, since the filter samples were tessellated into tetrahedra and the edges across sample boundaries were discarded i.e. non-periodic networks were considered. It is also sensible that the average connectivities of filters sized [2,2,4] are larger than those of filters sized [1,1,4] due to the smaller ratio of surface area to volume. As for nodes identified with the mDTM, the average connectivities is approximately 5.45, which is close to 5.7, the value obtained through experiments by Witt as cited by Shire & O'Sullivan (2017).

Table 3: Virtual filter samples created by Shire (2014) considered in the research.

| Filter | $C_u$ | $\mu$ | $e$ | Size | No. of Particles | DTM Applied |
|---|---|---|---|---|---|---|
| F12dense111 | 1.2 | 0.0 | 0.558 | $1 \times 1 \times 1$ | 8262 | mDTM |
| F12dense114 | | | | $1 \times 1 \times 4$ | 33048 | wDTM |
| F12dense224 | | | | $2 \times 2 \times 4$ | 132192 | wDTM |
| F12medium114 | | 0.1 | 0.651 | $1 \times 1 \times 4$ | 33048 | wDTM |
| F12medium224 | | | | $2 \times 2 \times 4$ | 132192 | wDTM |
| F12loose114 | | 0.3 | 0.714 | $1 \times 1 \times 4$ | 33048 | wDTM |
| F12loose224 | | | | $2 \times 2 \times 4$ | 132192 | wDTM |
| F15dense111 | 1.5 | 0.0 | 0.531 | $1 \times 1 \times 1$ | 9313 | mDTM |
| F15dense114 | | | | $1 \times 1 \times 4$ | 37252 | wDTM |
| F15dense224 | | | | $2 \times 2 \times 4$ | 149008 | wDTM |
| F15medium114 | | 0.1 | 0.608 | $1 \times 1 \times 4$ | 37252 | wDTM |
| F15medium224 | | | | $2 \times 2 \times 4$ | 149008 | wDTM |
| F15loose114 | | 0.3 | 0.658 | $1 \times 1 \times 4$ | 37252 | wDTM |
| F15loose224 | | | | $2 \times 2 \times 4$ | 149008 | wDTM |
| F20dense111 | 2.0 | 0.0 | 0.467 | $1 \times 1 \times 1$ | 12115 | mDTM |
| F20dense114 | | | | $1 \times 1 \times 4$ | 48460 | wDTM |
| F20dense224 | | | | $2 \times 2 \times 4$ | 193840 | wDTM |
| F20medium114 | | 0.1 | 0.523 | $1 \times 1 \times 4$ | 48460 | wDTM |
| F20medium224 | | | | $2 \times 2 \times 4$ | 193840 | wDTM |
| F20loose114 | | 0.3 | 0.555 | $1 \times 1 \times 4$ | 48460 | wDTM |
| F20loose224 | | | | $2 \times 2 \times 4$ | 193840 | wDTM |
| F30dense111 | 3.0 | 0.0 | 0.382 | $1 \times 1 \times 1$ | 22600 | mDTM |
| F30dense114 | | | | $1 \times 1 \times 4$ | 90400 | wDTM |
| F30medium114 | | 0.1 | 0.426 | $1 \times 1 \times 4$ | 90400 | wDTM |
| F30loose114 | | 0.3 | 0.455 | $1 \times 1 \times 4$ | 90400 | wDTM |

Table 4: Properties related to the void space topology of the virtual filter samples considered (values in brackets correspond to the mDTM).

| Filter | No. of Nodes/Pores | No. of Edges/Constrictions | Average Connectivity |
|---|---|---|---|
| F12dense111 | (9822) | (26666) | (5.43) |
| F12dense114 | 88418 | 171991 | 3.90 |
| F12dense224 | 380576 | 750019 | 3.94 |
| F12medium114 | 90400 | 175938 | 3.90 |
| F12medium224 | 388493 | 765685 | 3.94 |
| F12loose114 | 90568 | 176202 | 3.90 |
| F12loose224 | 389908 | 768434 | 3.94 |
| F15dense111 | (10693) | (29378) | (5.49) |
| F15dense114 | 99057 | 192950 | 3.90 |
| F15dense224 | 423302 | 834216 | 3.94 |
| F15medium114 | 100520 | 195821 | 3.90 |
| F15medium224 | 431374 | 850266 | 3.94 |
| F15loose114 | 100899 | 196538 | 3.90 |
| F15loose224 | 432929 | 853262 | 3.94 |
| F20dense111 | (13937) | (37790) | (5.42) |
| F20dense114 | 125630 | 244991 | 3.90 |
| F20dense224 | 534309 | 1053922 | 3.95 |
| F20medium114 | 127207 | 248049 | 3.90 |
| F20medium224 | 540819 | 1066849 | 3.95 |
| F20loose114 | 128032 | 249739 | 3.90 |
| F20loose224 | 542605 | 1070344 | 3.95 |
| F30dense111 | (25231) | (69301) | (5.49) |
| F30dense114 | 231730 | 453881 | 3.92 |
| F30medium114 | 234905 | 460027 | 3.92 |
| F30loose114 | 235923 | 461979 | 3.92 |

An algorithm was implemented to segment filter samples of sizes {[1,1,1], [1,1,2], [1,1,3]} and {[2,2,1], [2,2,2], [2,2,3]} from those with sizes of [1,1,4] and [2,2,4] respectively. To do so, the following steps were followed: firstly, the coordinates of the boundaries corresponding to the required size were evaluated. Secondly, the nodes outside these boundaries and all edges connected to these nodes were removed. Thirdly, the nodes at the boundary, defined as those that lose any number of edges during the second step, were assigned as inlets or outlets according to the flow direction.

*3.1.2. Geometry*

To allow the simulation of particle straining as illustrated in Figure 8(a), constriction sizes were required as constrictions form the narrowest portions of void spaces. A flow chart of the algorithm implemented with reference to Reboul et al. (2010) and Shire (2014) to obtain the constriction sizes is shown in Figure 12. As listed in Table 2, the three particles forming each constriction on the faces of tetrahedra were identified with DTM. Given the radius and the coordinates of the centroids of these particles, the radius and the coordinates of the centroid of the inscribed circle were calculated with Equations 1 and 2 in Reboul et al. (2010). However, inscribed circles identified this way were not necessarily part of the void space as it was possible that they coincide with other neighbouring particles. Therefore, any neighbouring particle that overlapped the inscribed circle was identified for each constriction. Using a combinatory approach, the properties of the inscribed circle were recalculated. This algorithm was validated in Appendix A.

On the other hand, the reduced cross-sectional area of each edge expressed as a fraction of the total area ($\beta_{ij}$ in Equation 10) is required to account for the effect of edge clogging on fluid flows as shown Figure 20. A flow chart of the algorithm implemented to obtain the reduced areas is presented in Figure 13.

It is important to understand how filter characteristics, such as $C_u$ and $e$, and the type of DTM applied affect the geometric properties of the void space. Therefore, the cumulative density functions of relevant geometric properties (listed in Table 6) for filters with size [1,1,1] were plotted; these filter samples were scaled such that the smallest filter particle has a diameter of $D_{0F} = 1$. With reference to Figures 14 and 15, the following observations were made:

1. An increase in filter $C_u$ implies that there is an increase in the number of larger particles considering the scaling applied. Hence, the formation of larger pores is expected, resulting in the increase of all geometric properties. This is also supported by the fact that the ratio of reduced edge area to total edge area remains relatively constant, since both the denominator and the numerator increase by similar proportions.

2. When the filter $e$ is reduced by reducing $\mu$, reversed effects can be observed i.e. the magnitude of all geometric properties decreases.

3. Node surface area is not affected by the filter density as samples of the same $C_u$ were created from the same cloud of non-contacting particles.

4. When the mDTM is applied, pores are merged as shown in Figure 9. Thus, larger node volume and surface area, as well as edge length, are observed. In addition, distributions of both edge area and diameter become steeper since larger edges connecting the merged pores are discarded.

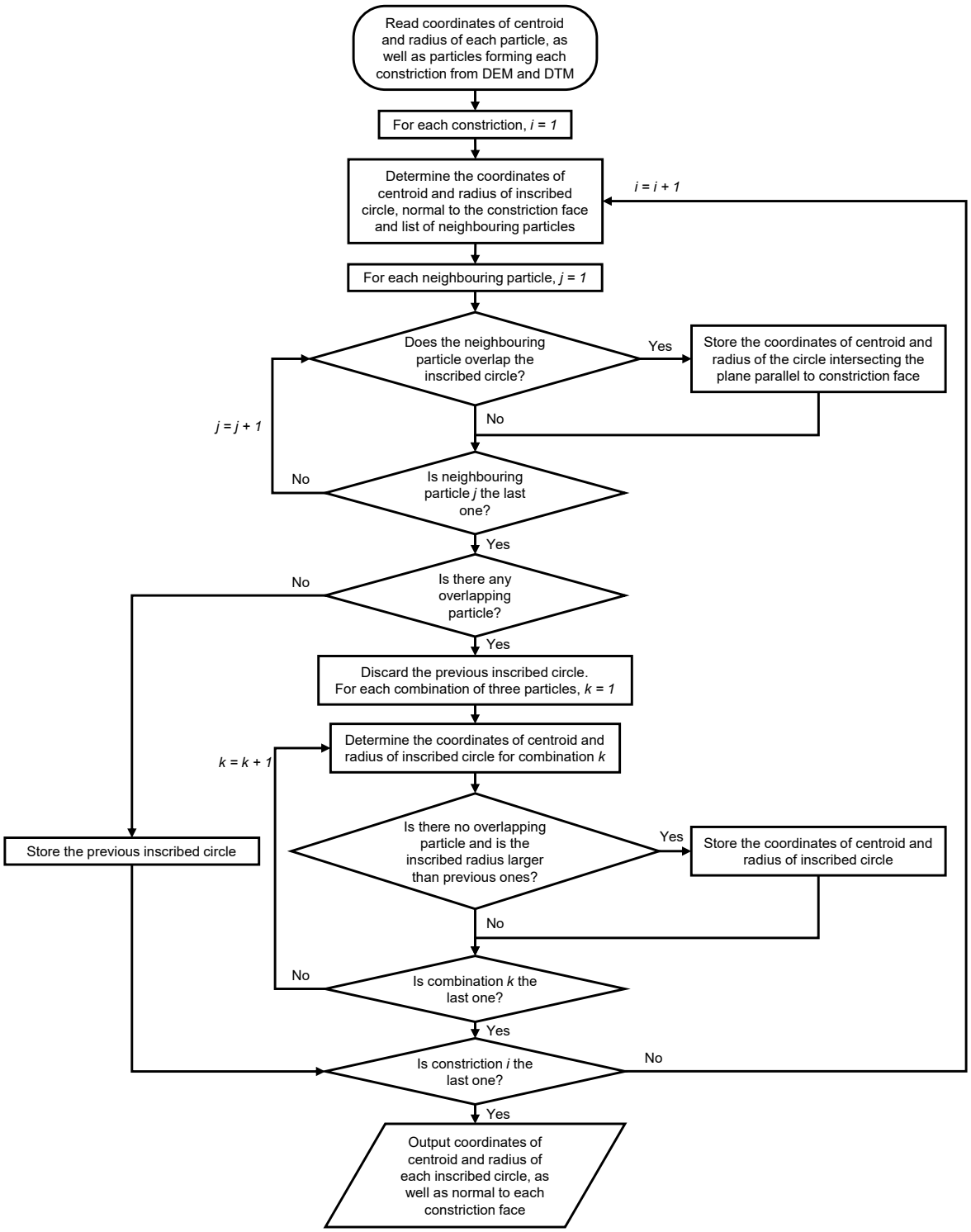



Figure 12: Flow chart of the algorithm used to obtain constriction size of each edge.

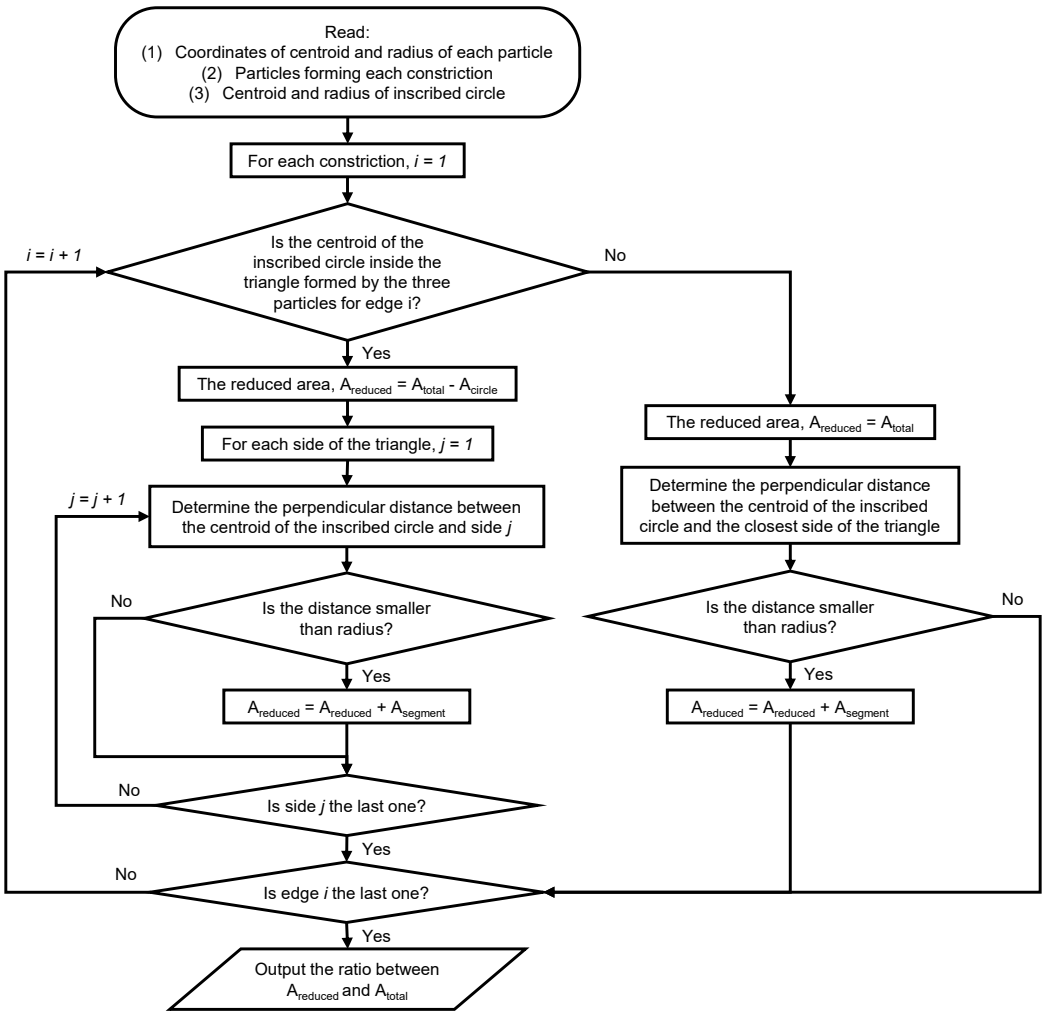
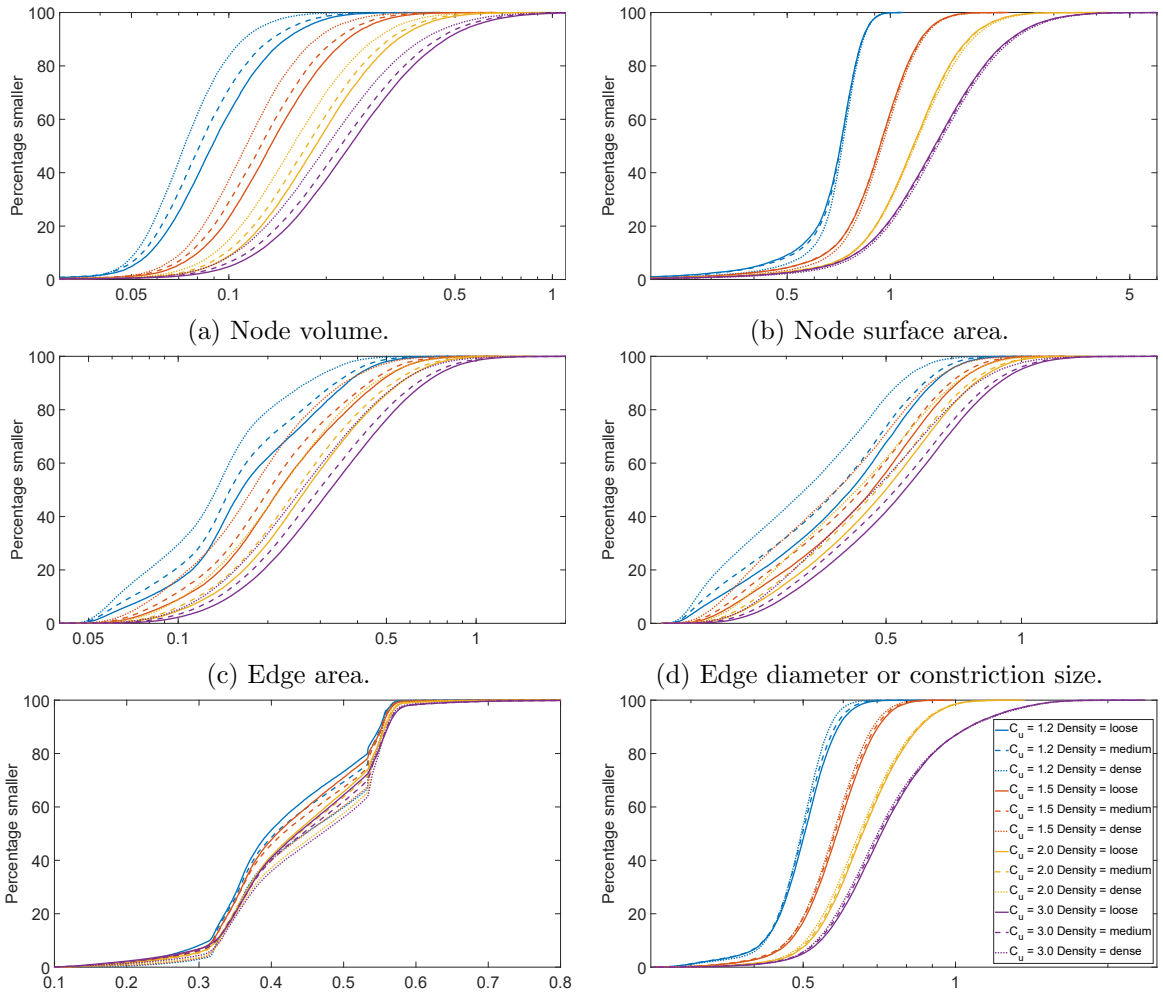



Figure 13: Flow chart of the algorithm used to obtain the ratio of the reduced area to the total area.




(a) Node volume.

(b) Node surface area.

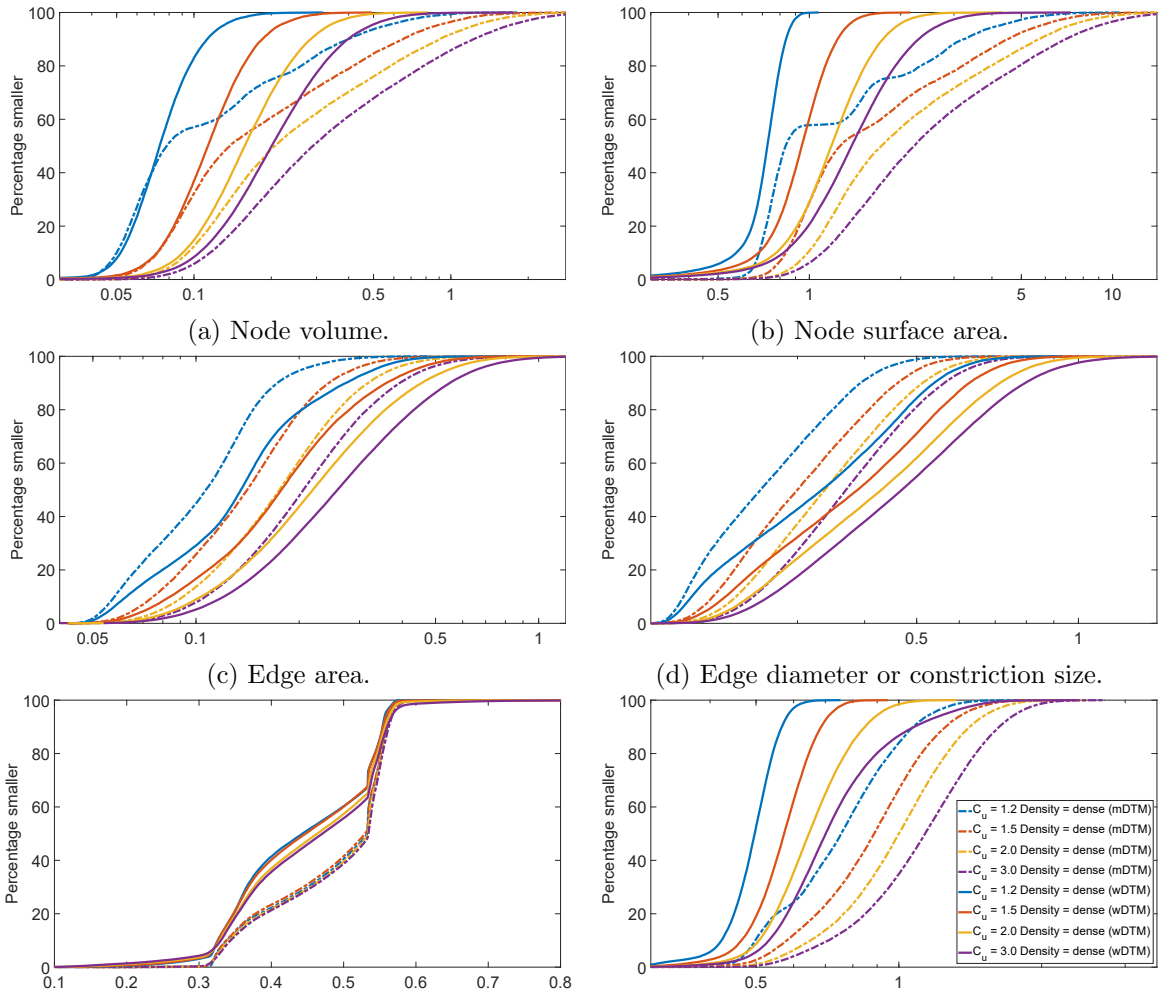
(c) Edge area.

(d) Edge diameter or constriction size.

(e) Reduced edge area as a fraction.

(f) Edge length.

Figure 14: Distributions of geometric properties of void spaces quantified with the wDTM.




(a) Node volume.

(b) Node surface area.

(c) Edge area.

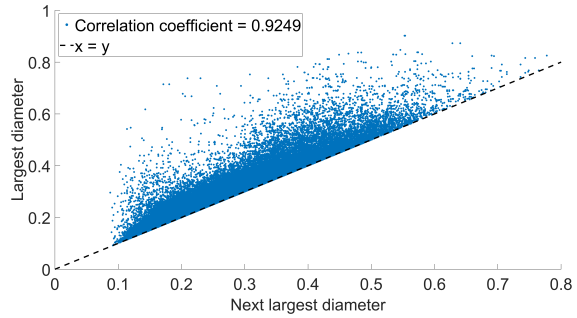
(d) Edge diameter or constriction size.

(e) Reduced edge area as a fraction.

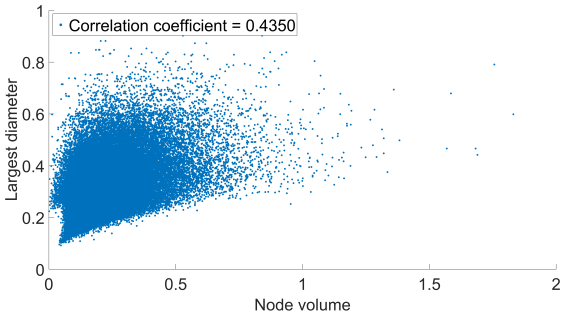
(f) Edge length.

Figure 15: Distributions of geometric properties of void spaces quantified with the mDTM.

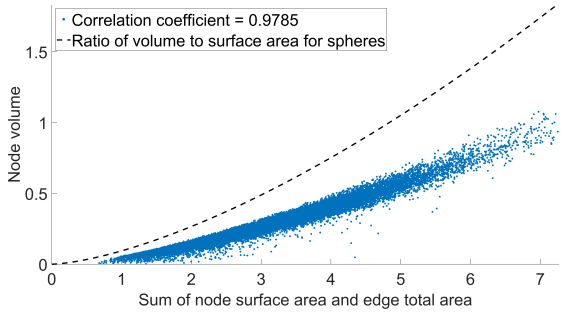
In their research to predict permeabilities in random monodisperse sphere packings, Bryant et al. (1993) showed that spatial correlation exists between void space geometric properties of neighbouring components. To demonstrate this, a similar process was followed, i.e. geometric properties were plotted against each other, as shown in Figure 16. In random sphere packings, larger pores are formed due to particles surrounding the pores being large or far from each other. Therefore, positive correlation between pore volume and constriction size is expected as shown in (b). Subsequently, the largest constriction size connected to each pore is also expected to be strongly correlated to the next largest one as shown in (a). Figures (c) to (f) are further discussed in the following section. These figures again emphasise that caution must be given when assuming geometric properties are randomly distributed during the construction of network models.



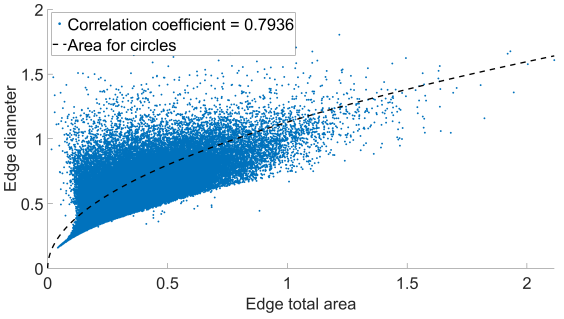

(a) Largest edge diameter against next largest.



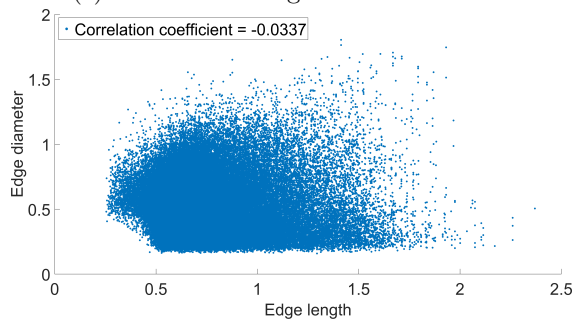

(b) Largest edge diameter against node volume.



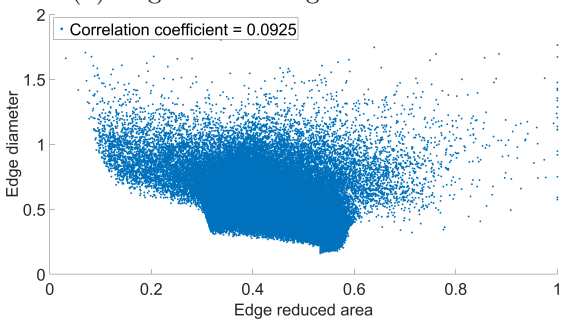

(c) Node volume against surface area.




(d) Edge diameter against total area.




(e) Edge diameter against edge length.




(f) Edge diameter against edge reduced area.

Figure 16: Spatial correlations in geometric properties for filter F30dense111(wDTM) ((a) and (e) adapted from Bryant et al. (1993)).

### *3.2. Network Generation*

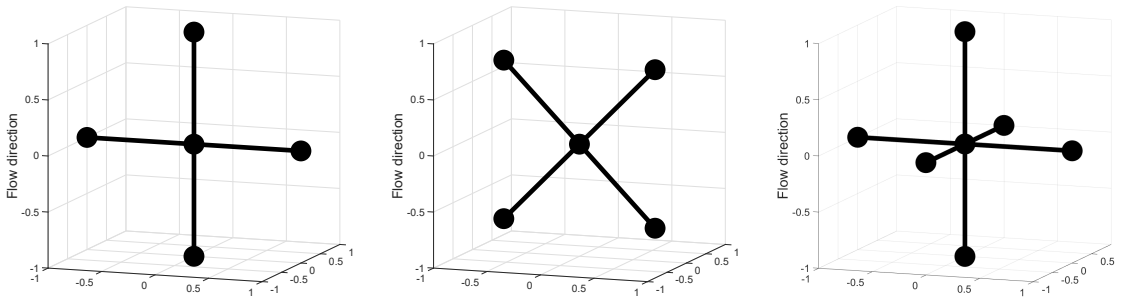
#### *3.2.1. Network Types*

To evaluate the significance of each component of void space structure identified in Figure 6, networks that match the actual topology and geometry of the void space to varying

extents were generated. With reference to Table 5, three types of networks were considered. The first type of networks was constructed based on regular lattices. Six regular lattices with varying degree of connectivity were considered in the research as shown in Figure 17. These include 2D networks such as simple square and slanted square, as well as 3D networks such as simple cubic (SC), slanted cubic, body-centred cubic (BCC) and face-centred cubic (FCC). The second and third types of networks were constructed based on the actual topology of the void space, i.e. the nodes were defined at the centroids of the pores and the edges represented the constrictions at the faces of the tetrahedra.

Table 5: Components of actual structure that are incorporated in each network type.

| Type | Network Description | Topology | Distribution | Spatial Correlation |
|---|---|---|---|---|
| 1 | Lattice topology with randomly assigned geometry | ✗ | ✓ | ✗ |
| 2 | Actual topology with randomly assigned geometry | ✓ | ✓ | ✗ |
| 3 | Actual topology with actual geometry | ✓ | ✓ | ✓ |




(a) Simple square (4). (b) Slanted square (4). (c) Simple cubic (6).

(d) Slanted cubic (6). (e) Body-centred cubic (8). (f) Face-centred cubic (12).

Figure 17: Regular lattices with varying degree of connectivity.

The geometric properties that were assigned to the nodes and edges of the networks are listed in Table 6. These properties were required to calculate fluid flow along each edge and to simulate particle straining. Assigning representative pore volumes to the nodes in the network allowed the simulation of node clogging as discussed in Section 2.3. Since the base particles were assumed to be spherical, a node was considered full when the total volume of base particles within the node reached 50% of the node volume. This is sensible given that uniform and random equal sphere packings have maximum theoretical packing densities of approximately 0.74 and 0.64 respectively.

Table 6: Geometric properties that are assigned to nodes and edges.

| Component | Geometric Properties |
|---|---|
| Node | Volume, surface area |
| Edge | Total cross-sectional area, reduced cross-sectional area, constriction size, length |

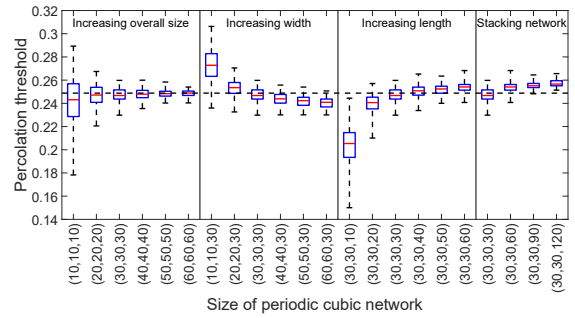
Two methods of assigning geometric properties were implemented. The first and second types of networks adopted the first method, in which geometric properties were randomly distributed in the system matching the probabilistic distributions obtained from virtual filter samples. As shown in Figure 16 (c) and (d), there are high correlations between node volume and surface area, as well as between edge total cross-sectional area and constriction size. Therefore, each pair of these properties was assigned using the same random value of cumulative density. As for edge reduced cross-sectional area and length, the median values were assigned since there were no obvious correlation as shown in Figure 16 (e) and (f). This implies that the properties of the nodes and edges in these types of networks are independent. The geometric properties in the third type of networks on the other hand were assigned based on the actual geometry of the void space, thus any spatial correlation that exists between these properties is taken into account implicitly as shown in Table 5.

*3.2.2. Percolation Threshold*

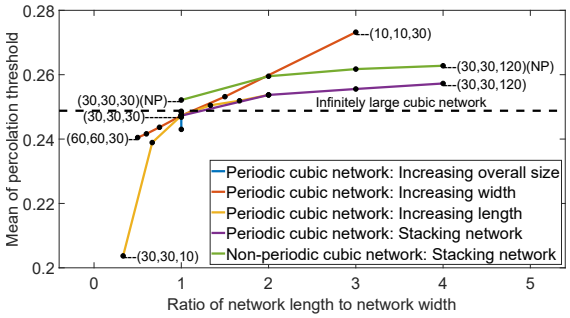
A network percolates when a cluster that spans between a face and the opposite face forms, i.e. a continuous path exists between the inlets and the outlets of the network. In the context of filtration, a continuous path is made up of 'open' edges that are larger than the flowing particle. The minimum percentage of the edges that must be open for the network to percolate is termed percolation threshold.

Periodic cubic networks of varying sizes were generated. The corresponding percolation thresholds were then determined by randomly 'opening' edges until the networks percolated. This process was repeated 150 times for each network such that the mean, median, range and standard deviation of percolation thresholds could be obtained. As illustrated with boxplots in Figure 18(a), the median of the percolation threshold increased to approach the theoretical value when the overall size was increased. On the other hand, when the width or the length of the periodic cubic networks were varied, the mean percolation threshold is shown in (a) and (b) to correlate positively to the network length to width ratio. Non-periodic cubic networks were also generated; these networks were shown to have larger percolation threshold than the theoretical value and their periodic counterpart in (b). This is sensible given that additional restrictions were imposed due to boundary effects.

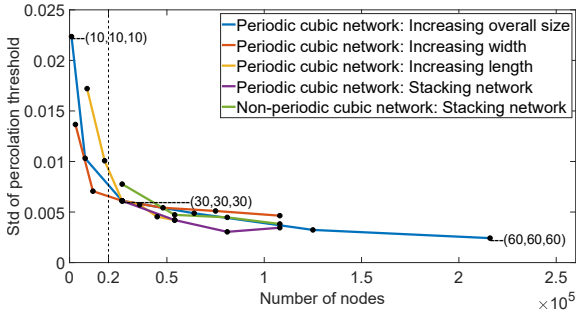
In (c), the standard deviation of the percolation threshold is demonstrated to be inversely proportional to the number of nodes. This suggests that sufficiently large networks, i.e. those with at least 20,000 nodes, should be used in the following simulations to produce representative results. It should be noted that the smallest network considered is filter F12dense111, which has 22,437 nodes. On the other hand, with reference to (d), the percolation thresholds of random sphere packings are approximately 51% as obtained by Mason and Meller Bryant et al. (1993); when the geometric properties were randomly distributed, the percolation thresholds reduced to approximately 39%. As for when the mDTM was applied, these values decreased significantly to 27% and 24% respectively.



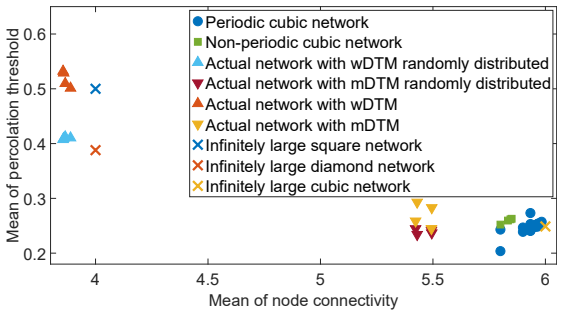

(a) Effects of network size.




(b) Effects of length to width ratio.




(c) Effects of node number.




(d) Effects of node connectivity.

Figure 18: Effects of various parameters on percolation threshold.

### *3.3. Fluid Flow Calculation*

When the filter void space is idealised as a network, the fluid flows in the overall system can be predicted at pore-scale. Calculating fluid flows in the network was an important step towards the simulation of filtration processes as the movements of suspended base particles are highly influenced by fluid motion; base particles move in the direction of fluid flow and are more likely to advance through constrictions with the larger flows.

#### *3.3.1. Local Conductance*

Fluid flows are usually relatively small in soils due to the low permeability. Therefore, laminar flows i.e. small Reynolds and Stokes numbers are expected (Trussell & Chang 1999). In such cases, fluid flows can be determined with the Stokes equation that describes momentum conservation at the microscopic scale (Chareyre et al. 2012):

$$\nabla p = \mu \nabla^2 u - \rho \nabla \Phi \tag{5}$$

where $p$ is the piezometric pressure (equal to the difference between the absolute pressure and the product of the density and the potential field), $\mu$ is the fluid viscosity, $u$ is the fluid velocity, $\rho$ is the fluid density, and $\Phi$ is the potential or the gravitational field. On the other hand, at the macroscopic scale, flows through porous materials are directly proportional to the pressure gradient, and are described with Darcy's equation:

$$q = -\frac{k}{\mu} \nabla p \tag{6}$$

where $q$ is the overall fluid flux per unit area and $k$ is the permeability. Both of the equations show a linear relationship between fluid flow and pressure difference, which is in line with laminar flows of incompressible fluids in tubes explained with Hagen-Poiseuille equation (Chareyre et al. 2012). The Hagen-Poiseuille equation has a coefficient

of proportionality termed the local conductance $g_{ij}$ as shown in Equation 7. For a circular tube with constant cross-sectional area, the local conductance is expressed as shown in Equation 8.

$$q_{ij} = g_{ij}(p_i - p_j) \tag{7}$$

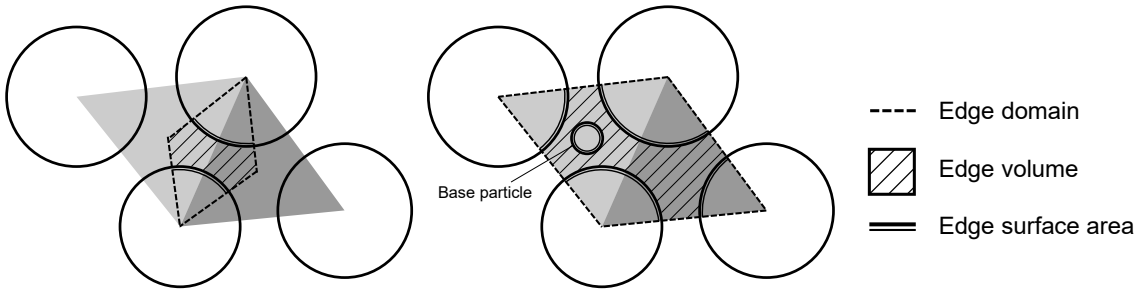
$$g_{ij} = \frac{\pi r_{ij}^4}{8\mu L_{ij}} \tag{8}$$

where $q_{ij}$, $g_{ij}$, $r_{ij}$ and $L_{ij}$ are the fluid flow, the local conductance, the radius and the length of edge $i \rightarrow j$ respectively, while $p_i$ is the local pressure at node $i$.

However, the constrictions in a filter's void space are non-circular and have varying cross-sectional areas along their axes. Therefore, the definition of local conductance based on hydraulic radius $R_{ij}^h$ proposed by Chareyre et al. (2012) was adopted, since they showed that the permeabilities through sphere packings predicted with this model are comparable to those obtained by solving Equation 5 with finite element method. The expression of the local conductance is given as:

$$g_{ij} = \alpha \frac{A_{ij}(R_{ij}^h)^2}{\mu L_{ij}}, R_{ij}^h = \frac{V_{ij}}{S_{ij}} \tag{9}$$

where $\alpha$ is the shape factor taken as 0.5, $A_{ij}$ is the total cross sectional area, while $S_{ij}$ and $V_{ij}$ are the surface area of the fluid-solid interface and the volume within the domain of edge $i \rightarrow j$ respectively. With reference to Figure 19(a), Chareyre et al. (2012) defined the domain of an edge as the union of the two tetrahedra formed from the contacting triangular face and the centroids of adjacent nodes.




(a) Original. (b) Modified.

Figure 19: 2D representation of conductance model (adapted from Chareyre et al. (2012)).

The presence of base particles within the filter void space also affects the fluid flow, hence two modifications were made to Equation 9. Firstly, the domain of an edge adopted in this research was defined as the union of the two adjacent tetrahedral void cells or nodes as shown in Figure 19(b). If any base particle entered the domain, the volume of the particle was deducted from $V_{ij}$ while the surface area of the particle was added to $S_{ij}$. Secondly, $A_{ij}$ was multiplied by the ratio of reduced area to the total area when edge $i \rightarrow j$ was clogged (Figure 8(a)). The modified expression is thus given by:

$$g_{ij} = \alpha \frac{(\beta_{ij} A_{ij})(R_{ij}^h)^2}{\mu L_{ij}}, R_{ij}^h = \frac{V_{ij}}{S_{ij}} \tag{10}$$

where $\beta_{ij}$ is equal to 1 when the edge is open, and is equal to the ratio of reduced area

to total area when the edge is clogged as calculated with the algorithm shown in Figure 13. An example showing the densest configuration of monodisperse particles, which has a value of $\beta_{ij}$ of 0.53, is given in Figure 20.

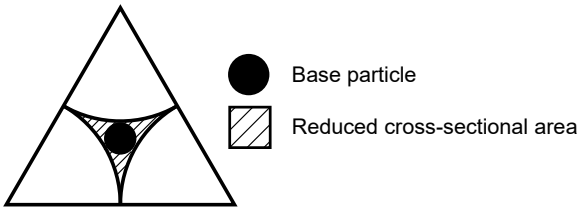



Figure 20: Reduction in cross-sectional area due to edge clogging.

*3.3.2. System of Equations*

The principle of conservation of mass is based on the simple concept that matter is neither created nor destroyed, and can be equally applied to solids as well as flowing fluids. If a control volume is defined at node $i$ of the network, conservation of mass requires that the total outflow to be equal to the source within the control volume. Assuming that the fluid is incompressible, the void space is fully saturated and the system is at steady state, the equation is given by:

$$\sum_{i \to j}^{c_i} q_{ij} = \sum_{i \to j}^{c_i} g_{ij}(p_i - p_j) = s_i \tag{11}$$

where $s_i$ and $c_i$ are the source and the connectivity of node $i$. For a network with $n$ number of nodes, $n$ simultaneous equations are formed. The system of equations can be expressed in matrix form:

$$\begin{bmatrix} G_{inlet,inlet} & G_{inlet,internal} & G_{inlet,outlet} \\ G_{internal,inlet} & G_{internal,internal} & G_{internal,outlet} \\ G_{outlet,inlet} & G_{outlet,internal} & G_{outlet,outlet} \end{bmatrix} \begin{bmatrix} P_{inlet} \\ P_{internal} \\ P_{outlet} \end{bmatrix} = \begin{bmatrix} S_{inlet} \\ S_{internal} \\ S_{outlet} \end{bmatrix} \tag{12}$$

where $[G]$ is the conductance matrix, $[P]$ is the local pressure vector and $[S]$ is the source vector. The source vector represents the external flow into each node. Note that inlets refer to the nodes along the boundary between the filter and the base material, whereas outlets refer to the nodes along the opposite boundary. Fluid flows through the filter from inlets to outlets. For internal nodes, $S_{internal}$ is zero. A Dirichlet boundary condition is imposed on the system of equations by specifying the local pressures at the inlets and outlets of the network, i.e. $P_{inlet}$ and $P_{outlet}$. This results in $n$ unknowns, namely $[P_{internal}]$, $[S_{inlet}]$ and $[S_{outlet}]$, which can be solved easily with Gaussian elimination. The solution gives the local node pressures at every node, which can be substituted into Equation 7 to determine the fluid flow in each edge. Lastly, a measure of permeability of the filter layer $k$ can be computed by rearranging Equation 6:

$$k = \frac{Q\mu L}{A(p_{inlet} - p_{outlet})}, Q = \sum [S_{inlet}] = -\sum [S_{outlet}] \tag{13}$$

where Q is the total flow, while L and A are the length and the cross-sectional area of the filter. Validations were performed by comparing the predicted values of permeabilities against other research in Appendix A.

*3.4. Simulation Methodology*

A flow-biased walk algorithm, developed with reference to Rege & Fogler (1987), Shire & O'Sullivan (2017) and Ding et al. (2018), was used to simulate filtration processes. The flow chart of the algorithm is shown in Figure 21, and the algorithm was validated in Appendix A. A brief description of the simulated filtration processes is as follows. As shown in Figure 1(a), granular filters are typically employed adjacent to the base material. Therefore, base particles that are dragged along by seepage flow enter the network through the inlets and move within the network until they are retained or eroded. Eroded base particles are those that have traversed the whole depth of the filter and subsequently leave the network through the outlets. Note that filtration occurs only through particle-straining.

Spherical base particles were introduced at the inlets of the network for the first $T_{intro}$ number of time steps, where the value of $T_{intro}$ was chosen to be five times the shortest span of the network in the direction of seepage. This is to ensure that large number of base particles were simulated such that the network model produced statistically significant and representative results. For each time step, new base particles were introduced based on a chosen PSD until the inlets, i.e. the nodes along the boundary between filter and base material, were full. Hence, the number of base particles introduced varied every time step. Furthermore, the node to which each base particle was introduced was randomised to capture the stochastic nature of particle movements.

As mentioned, base particles that were introduced flow within the network until they were retained or eroded. Thus, a base particle could be referred to as (1) 'active' if it was free to move in the direction of flow, (2) 'inactive' if it was retained due to filtration through particle straining, or (3) 'eroded' if it reached the outlets. Note that each base particle moved directly to a neighbouring node within a time step, irrespective of the flow velocity and the distance between centroids of nodes. The target node was determined based on flow-biased probabilities as proposed by Rege & Fogler (1987) (see Figure 7(b)). During each time step, the active base particles could do one of the following once the target node was determined (see Figure 22):

1. If all edges with outflow had been clogged, the particle was retained and became inactive.

2. If the particle was larger than the constriction leading to target node, the edge was clogged and the particle was retained and became inactive.

3. If the particle was smaller than the constriction but the target node was filled with inactive particles, the edge was clogged and the particle was retained and became inactive.

4. If the particle was smaller than the constriction but the target node was full (but not filled with inactive particles), the particle stayed in the same node while still being active.

5. Otherwise, the particle was moved to the target node.

Distinction had to be made between actions (3) and (4) as it was possible that the target node was temporarily full; the target node would have sufficient space in the subsequent time steps when active particles left. It is important to note that once an edge was clogged or a base particle became inactive, they were assumed to remain so until the end

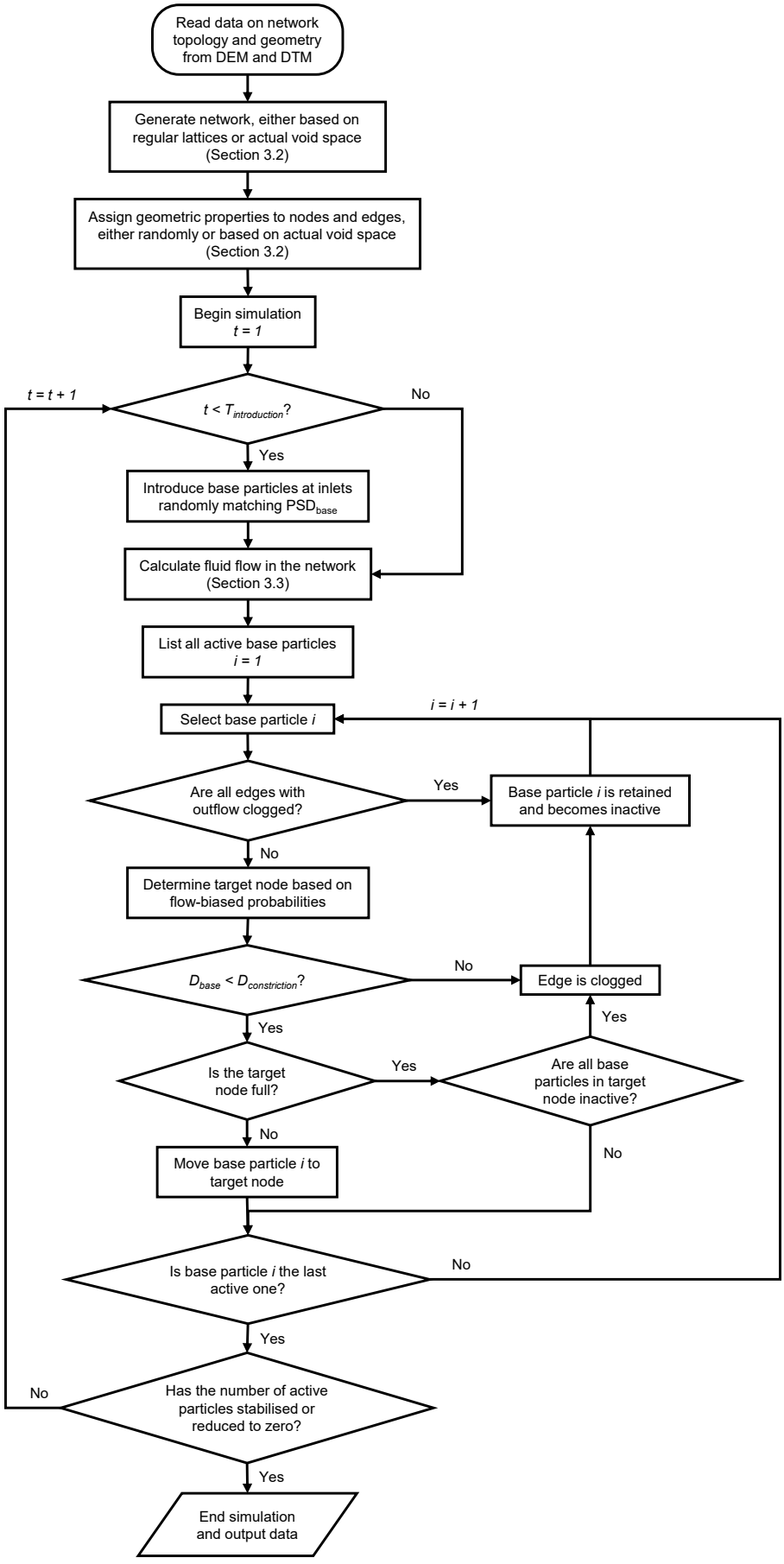



Figure 21: Flow chart of the simulation.

of simulation. In addition, edges do not have associated volumes; therefore, base particles could only be retained in nodes.

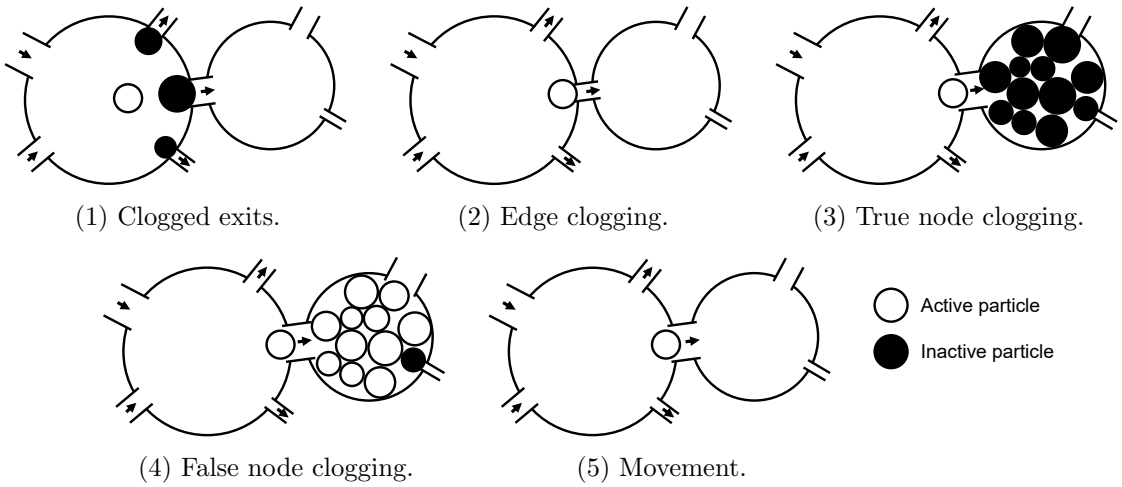



Figure 22: Possible actions for active base particles every time step (adapted from Shire & O'Sullivan (2017)).

Among the simulations that were performed, a small fraction exhibited a behaviour in which a few base particles moved between two nodes endlessly due to alternating flow direction in the edge connecting the nodes. This was attributed to the lack of a simulated time step and the positions of particles being discrete, such that the particles overshot the point where the flow was zero. These base particles were assumed to be retained. Simulations were ended once the number of active particles reduced to zero or stabilised.

## 4. Results and Discussions

### *4.1. Simulations with Monodisperse Base Particles*

Simulations with monodisperse base particles reduced the number of variables, thereby enabling a basic understanding of the filtration processes. For a given filter, multiple independent monodisperse simulations were performed with the normalised diameter of base particle $d^n$ increasing from 0.15 (just below the minimum constriction size) with an increment of 0.005. The upper limit of $d^n$ was chosen such that the percentage of base particles that were eroded by volume $p_{eroded}$ had reduced to 0%. The repeatability of these simulations was proven in Appendix A. Three types of simulations were performed:

1. More base particles were introduced, i.e. particles were introduced for $T_{intro}$ equal to five times the shortest span of the network in the direction of seepage as discussed in Section 3.4. (Sections 4.1.2 and 4.1.4)

2. Continuous filtration, i.e. ten simulations of the first type were performed consecutively. (Section 4.1.3)

3. Less base particles were introduced, i.e. $T_{intro} = 5$, so that simulation time was reduced for larger networks. (Section 4.1.5)

#### *4.1.1. Comparison of Largest Erodible Particles*

To ensure that the adopted network model produced reasonable results, the normalised diameters of the largest erodible particle $d^n_{largest}$ were determined and compared with those obtained in other research as shown in Figure 23. In their hydraulic and dry-vibration filter tests, Kenney et al. (1985) discovered that $d^n_{largest}$ were 0.18 and 0.25 when using filters with $C_u$ of 1.2 and 3.0 and $e$ within ranges of 0.52-0.56 and 0.42-0.47 respectively. Therefore, network model simulations were carried out using filters F12dense* and F30loose* since these filters have comparable $e$ of 0.558 and 0.455 respectively. Figure 23 shows that the adopted network model was able to capture the trend, i.e. $p_{eroded}$ reduces significantly at a similar value of $d^n$. However, the adopted network model consistently over-predicted the values of $d^n_{largest}$. On the other hand, the value of $d^n_{largest} = 0.29$ (see Section 4.1.4) for filters F12dense* agrees well with that obtained by Roozbahani et al. (2014) at 0.27, who simulated filtration processes using geometrical DEM with monodisperse spherical particles.

The discrepancies between the predicted values and the experimental results (Kenney et al. 1985) were attributed to three possible factors. Firstly, the assumption of spherical filter and base particles may not be valid since the CSD may be affected as discussed in Section 2.4 (Taylor 2016). Secondly, given that the network model was able to simulate particle straining only, other filtration mechanisms such as friction may play significant roles, though Kenney et al. (1985) minimised the possibility of particle bridging by introducing large hydraulic gradients and applying regular gentle tapping. Lastly, the ratio of number of base particles introduced to network length in the direction of seepage is shown to affect filter effectiveness in Section 4.1.3.

#### *4.1.2. Effects of Network Type*

As proposed in Section 3.2, the three different types of networks outlined in Table 5 were utilised in filtration simulations for filter F12dense111 to examine the significance of accurate description of the individual components of void space structure. These networks

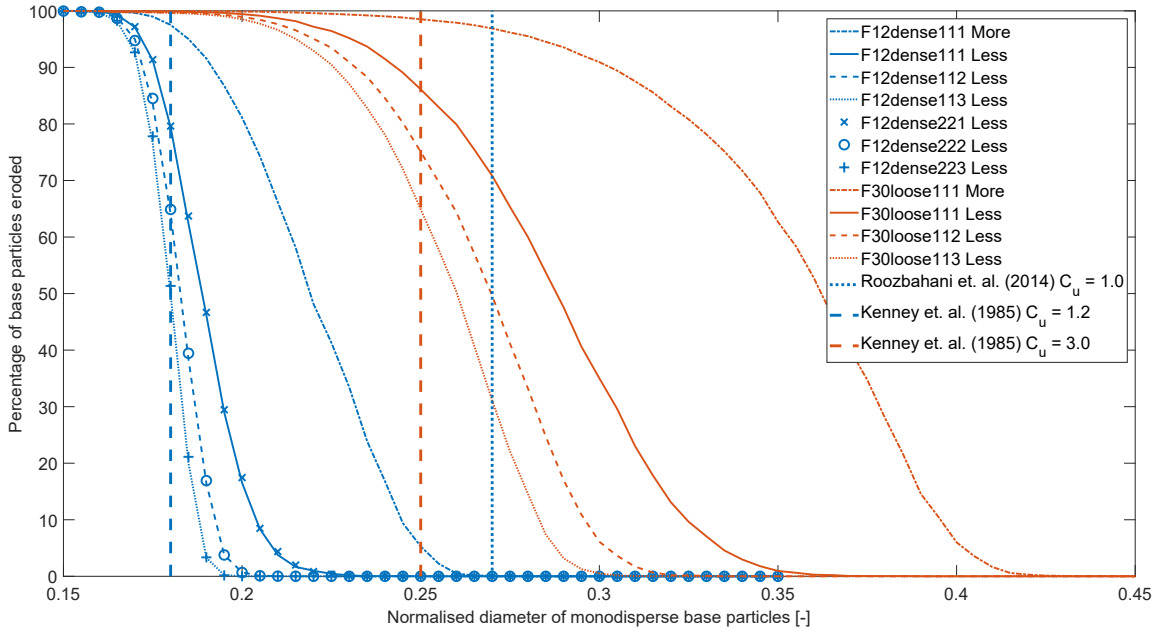
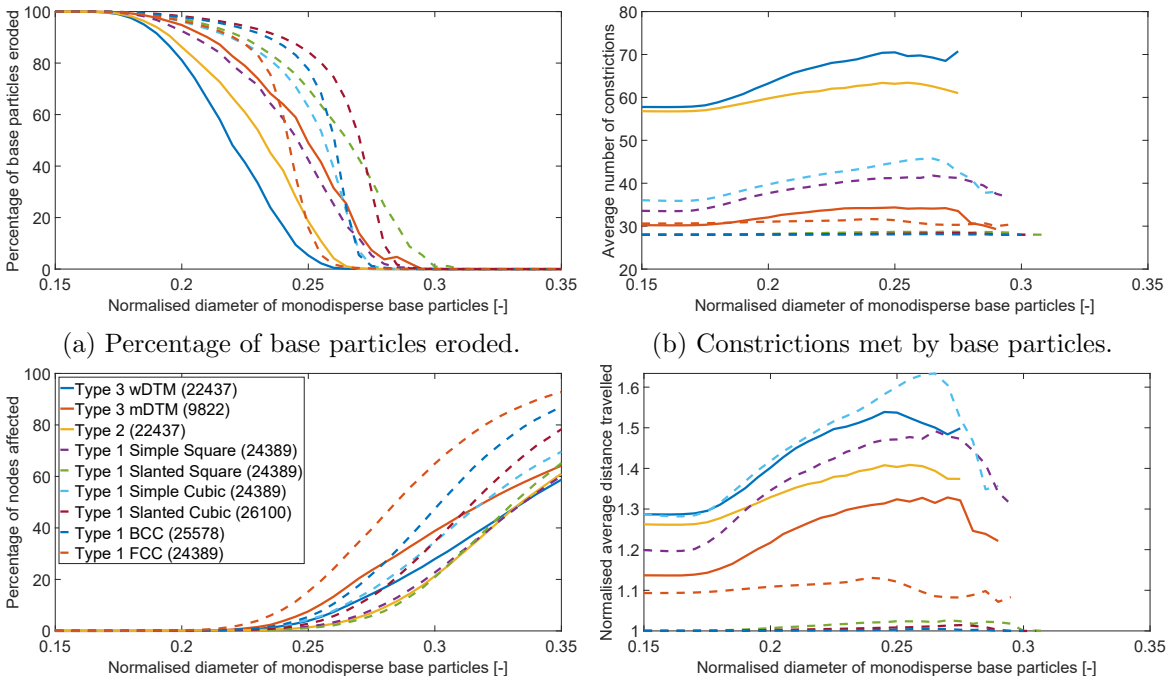



Figure 23: Comparison of largest particle eroded against Kenney et al. (1985) and Roozbahani et al. (2014); 'More' and 'Less' represent the number of base particles simulated.




(a) Percentage of base particles eroded.

(b) Constrictions met by base particles.

(c) Nodes affected by premature node clogging.

(d) Path tortuousity.

Figure 24: Comparison of different network types (Values in brackets represent the corresponding numbers of nodes).

have comparable number of nodes and non-periodic boundaries. Figure 24 clearly shows that filtration characteristics are highly dependent on the type of network model.

The concept of percolation threshold discussed in Section 3.2.2 is revisited here. With reference to Table 7, percolation thresholds of the networks considered were first determined using undirected networks. The obtained percolation thresholds then allowed the evaluation of the expected $d^n_{largest}$, which is defined as the corresponding diameter on the constriction size (or edge diameter) distribution at a percentage of 100 minus percolation threshold. However, these $d^n_{largest}$ have magnitudes that are significantly larger than those observed in Figure 24(a). This is because the use of undirected networks did not consider the fluid flows, which were assumed to dictate particle movements. Therefore, the values of percolation threshold were re-evaluated using directed networks. This resulted in an increase in percolation threshold, thus a reduction in $d^n_{largest}$, as particles were forced to move in the direction of flows. The predicted $d^n_{largest} = 0.293$ for network type 3 agrees well with that obtained through simulations at 0.29 (see Figure 28). However, the difference in the predicted and simulated values of $d^n_{largest}$ increases as connectivity increases.

Table 7: Percolation thresholds, as well as the ratio of the number of edges leading to the same layer to the number of edges leading to the next layer, for the networks considered.

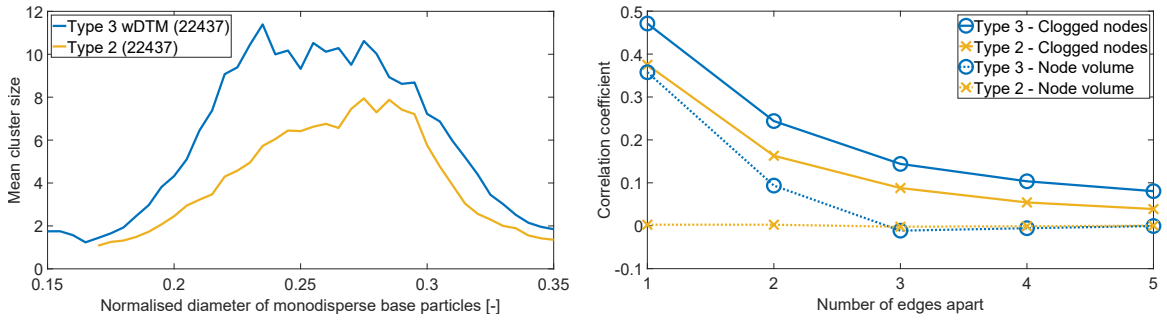
| Network | Undirected | $d^n_{largest}$ | Directed | $d^n_{largest}$ | Ratio |
|---|---|---|---|---|---|
| Type 1 Simple Square | 0.4080 | 0.376 | 0.4337 | 0.364 | 2 |
| Type 1 Slanted Square | 0.3672 | 0.395 | 0.3672 | 0.395 | 0 |
| Type 1 Simple Cubic | 0.2546 | 0.450 | 0.3184 | 0.419 | 4 |
| Type 1 Slanted Cubic | 0.2293 | 0.462 | 0.2683 | 0.443 | 0 |
| Type 1 BCC | 0.1862 | 0.483 | 0.2439 | 0.455 | 0 |
| Type 1 FCC | 0.1255 | 0.520 | 0.1502 | 0.504 | 1 |
| Type 2 | 0.4149 | 0.373 | 0.5106 | 0.328 | - |
| Type 3 wDTM | 0.5479 | 0.311 | 0.5879 | 0.293 | - |
| Type 3 mDTM | 0.2956 | 0.314 | 0.3222 | 0.306 | - |

Although the concept of percolation threshold is a useful tool in explaining filtration processes, it is only able to capture the influence of network connectivity and CSD. Another factor affecting the filter effectiveness is the assignment of representative volume to network nodes, which enabled the simulation of premature node clogging. Premature node clogging occurs when a node becomes full before all exits are clogged, i.e. the node volume is smaller than the total volume of (n+1) number of base particles, where n is the number of exits connected to the node that are smaller than the simulated $d^n$. To evaluate the value of $d^n$ at which node volume dominates filtration processes, the percentages of nodes affected by premature node clogging were plotted against $d^n$ in Figure 24(c). The figure shows that premature node clogging in networks with higher connectivity occurred at smaller $d^n$. Furthermore, the values of $d^n$ at which the percentage of nodes affected increases correspond well with those at which $p_{eroded}$ reduces in Figure 24(a), e.g. at $d^n$ of around 0.25, 0.27 and 0.28 for FCC, BCC and Slanted Cubic networks respectively.

It is demonstrated that when the concepts of percolation threshold and premature node clogging are considered in conjunction, the values of $d^n$ at which each filter became effective, i.e. $p_{eroded}$ dropping to zero, can be fully explained. Both of these properties are directly controlled by the void space structure, highlighting that assuming lattice topology can produce misleading results.

Networks of type 2 were studied to emphasise the significance of spatial correlation in simulating filtration processes. In Figure 24(a), filters based on networks of type 3, i.e. networks with spatial correlation implicitly accounted for, were shown to be more effective than those based on networks of type 2. This is because the existence of spatial correlation led to not only an increase in percolation threshold (see Table 7), but also a higher percentage of nodes affected by premature node clogging (see Figure 24(c)). The higher percentage is caused by the larger number of exits smaller than $d^n$ for the affected nodes, since the edges connected to each node are similarly sized in real packings as proven in Figure 16(a).

To further understand how incorporating spatial correlations influenced filtration processes, the clusters of clogged nodes at the end of simulations were analysed. With reference to Figure 25(a), the mean size of these clusters in network type 3 is larger than those in network type 2. Visualisation is also given in Figure 26, which contains plots of the ten largest clusters. This suggests that nodes adjacent to fully-filled nodes are more likely to clog for network type 3. To prove this numerically, a two-point correlation analysis was performed on the nodes that were clogged at the end of simulations. As illustrated in Figure 25(b), the correlation coefficients of the boolean values representing whether each node was clogged were calculated for distances of one to five number of edges. Both types of network exhibited positive close-range correlation as expected because nodes that are adjacent to clogged nodes would have less 'open' edges connected. However, consistently higher correlation was observed in network type 3, specifically 0.4712 as opposed to 0.3750 for a distance = 1. The difference is attributed to the nodes within close proximity having similar volumes for network type 3 as shown in Figure 25(b). Therefore, since the volume of each node is correlated to the constriction sizes of the connected edges (see Figure 16(b)), it is more probable to encounter, in network type 3, groups of smaller nodes with smaller connected edges that are more susceptible to clogging.

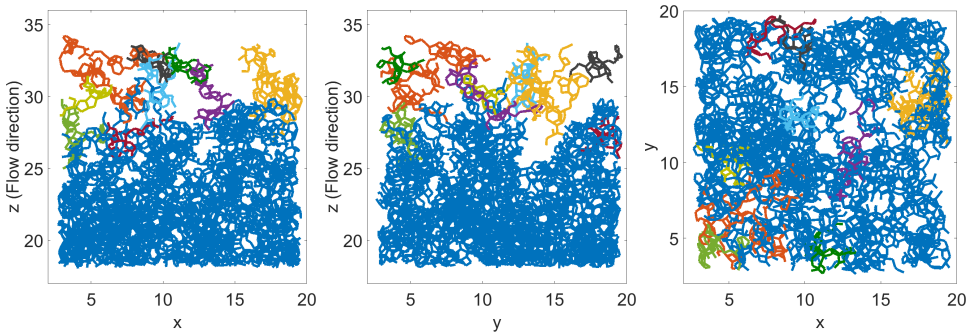



(a) Average size of clusters of clogged nodes. (b) Two-point spatial correlation for $d^n = 0.235$.

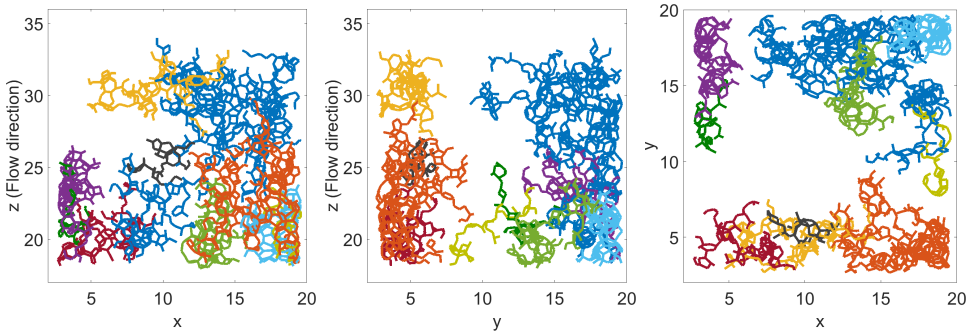
Figure 25: Significance of incorporating spatial correlation in simulating filtration processes.

The paths taken by the eroded particles were also examined. The use of lattice topology, i.e. network type 1, introduced a manner of regularity in the arrangement of nodes and edges that does not exist in random packings. Therefore, even though networks of comparable number of nodes were considered, the number of constrictions passed through by the eroded particles reduced significantly as shown in Figure 24(b). It is interesting to note that the number of constrictions was also influenced by the ratio of the number of exits leading to the same layer to the number of exits leading to the next layer; the higher the ratio, the larger the number of constrictions (see Table 7).

A measure of path tortuousity, defined as the average distance travelled by the eroded particles normalised by the shortest path, is plotted in Figure 24(d). The use of network




(a) Network type 3.




(b) Network type 2.

Figure 26: 10 largest clusters of nodes that were clogged for networks of types 2 and 3.

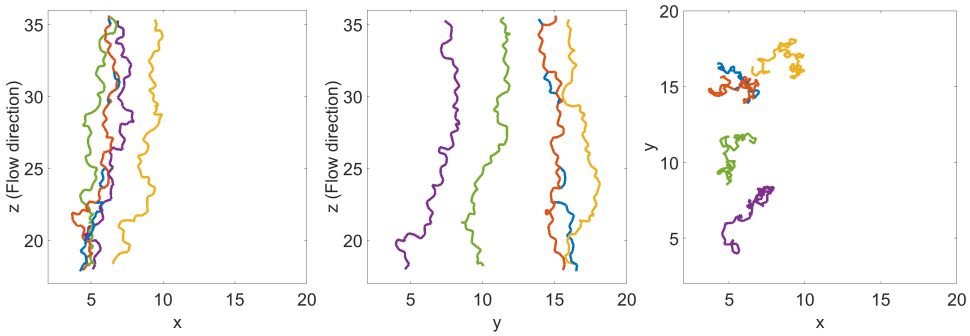
type 3 resulted in higher path tortuousity when compared to that of network type 2. This could be due to the base particles having to travel around larger and more connected clusters of clogged nodes that are observed in network type 3. In addition, it can be inferred that the path tortuousity is a function of both $d^n$ and simulation time. For smaller $d^n$, a higher proportion of edges is larger than the base particles; thus, particles were able to traverse the networks directly. On the other hand, base particles with larger $d^n$ were required to avoid smaller edges, resulting in more tortuous paths. Furthermore, the number of edges and nodes that are clogged, and thus the path tortuousity, increases with simulation time as shown in Table 8. The paths taken by the eroded particles at small and large simulation time are also plotted for comparison in Figure 27.

Table 8: Evolution of path tortuousity with simulation time for network type 3 with $d^n = 0.245$.

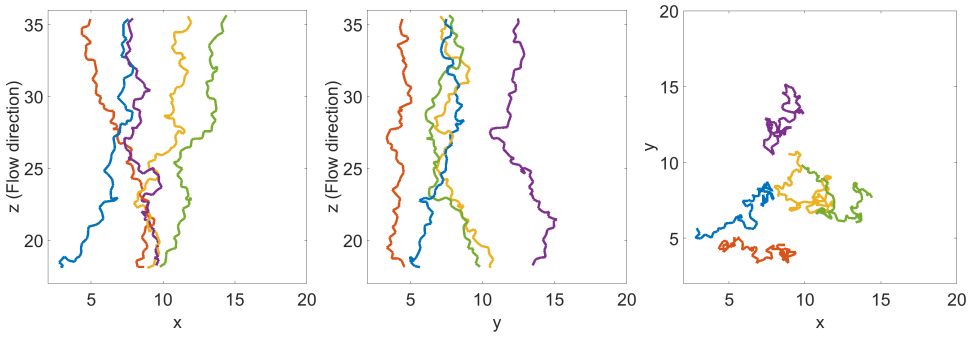
| Simulation Part | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Path Tortuousity | 1.422 | 1.495 | 1.517 | 1.530 | 1.537 |

Simulations using networks constructed based on the wDTM and the mDTM gave comparable results. As shown in Table 12, the predicted $d^n_{largest}$ for the networks are similar even though percolation thresholds dropped significantly when considering the mDTM, since edges with larger diameters were discarded when nodes were merged in the mDTM (see Section 3.1.2). This is also reflected in the simulations, in which $d^n_{largest} \approx 0.295$ for both networks. However, the disadvantages associated with the use of the mDTM include:

1. Additional computation, in particular to determine inscribed spheres that are within void space (see Sufian et al. (2015) Appendix 1), is required.




(a) Small simulation time.




(b) Large simulation time.

Figure 27: Paths taken by the eroded particles at different simulation time for network type 3 with $d^n = 0.245$.

2. The assignment of a volume to each node is less meaningful. Consider a larger yet elongated node merged from two tetrahedral cells; it is possible for the whole node to become full when only the edges connected to one of the cells are clogged.

3. Given that the length of each edge was calculated based on the centroids of the connected nodes, the calculated distance travelled by particles may not be valid, i.e. based on paths that are not part of the void space.

*4.1.3. Effects of Continuous Simulation*

Having established that it is important to accurately describe void space structure in network models, all subsequent simulations considered only network type 3 using the wDTM unless stated otherwise. In the previous section, filtration characteristics such as path tortuousity are shown to vary as more base particles were introduced. Therefore, ten consecutive simulations were performed at selected values of $d^n$ using filter F12dense111 to amplify the effects of simulation time. A similar framework for assessing the filter effectiveness proposed by Foster & Fell (2001) is adopted. With reference to Figure 28, these categories were observed in simulations with filter F12dense111 within ranges of $d^n$ listed in Table 9.

For $d^n \leqslant 0.155$, all base particles were smaller than the minimum constriction size of 0.159, and hence were expected to travel through the filter network without being retained. As $d^n$ was increased from 0.160 to 0.280, the number of constrictions that were smaller than the base particles increased as well. This implies that there were more sites available to capture base particles, thus $p_{eroded}$ reduced from 100% to 0%. It should be noted that $p_{eroded}$ was expected to increase with every consecutive simulation for filters characterised as partial seal with continuous erosion. This is because the available retaining sites became

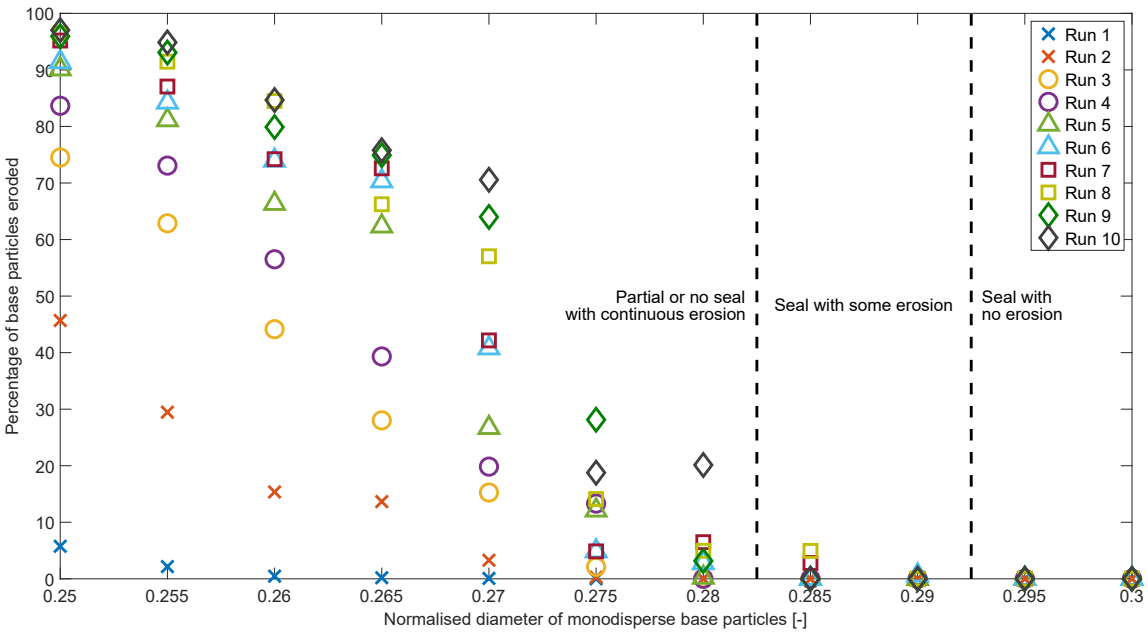



Figure 28: Percentages of base particles eroded in continuous simulations with F12dense111.

Table 9: Characterisation of filter effectiveness in monodisperse simulations (Foster & Fell 2001).

| Filter Effectiveness | Range of $d^n$ |
|---|---|
| No seal with continuous erosion | $d^n \leqslant 0.155$ |
| Partial seal with continuous erosion | $0.160 \leqslant d^n \leqslant 0.280$ |
| Seal with some erosion | $0.285 \leqslant d^n \leqslant 0.290$ |
| Seal with no erosion | $0.295 \leqslant d^n$ |

saturated as more base particles were introduced and retained with increasing simulation time. Eventually, a portion of the inlets became clogged, resulting in base particles being introduced only to the remaining inlets which are connected to erodible paths.

Three observations were made in regard to the erodible paths for specific $d^n$ values. Firstly, the number of erodible paths is affected by $d^n$, such that base particles with larger $d^n$ tend to follow smaller number of common paths. This can be seen when the paths followed by particles with $d^n$ of 0.250, 0.265 and 0.280 are plotted in Figure 29. Secondly, path tortuousity appears to increase with $d^n$, which is consistent with Figure 24(d). Thirdly, the erodible paths remained relatively similar for all ten consecutive runs, as demonstrated visually in Figure 30. A numerical study is performed to quantify the similarity of erodible paths. With reference to Sufian et al. (2019), a measure of similarity, S, comparing two sets of edge indices followed by particles that were eroded, {a} and {b}, can be defined as the number of similar edges divided by the average number of edges in both sets:

$$S = \frac{|\{a\} \cap \{b\}|}{0.5(|\{a\}| + |\{b\}|)} \tag{14}$$

The values of S for consecutive runs with $d^n$ ranging between 0.25 and 0.28 were evaluated and tabulated in Table 10. It should be noted that the missing values correspond to runs in which no base particle was eroded. Since most values of S are close to one, the similarity of erodible paths with simulation time is confirmed. This is especially true for runs 6 to 10, i.e. when the filter networks had stabilised and most of the available retaining sites became saturated.

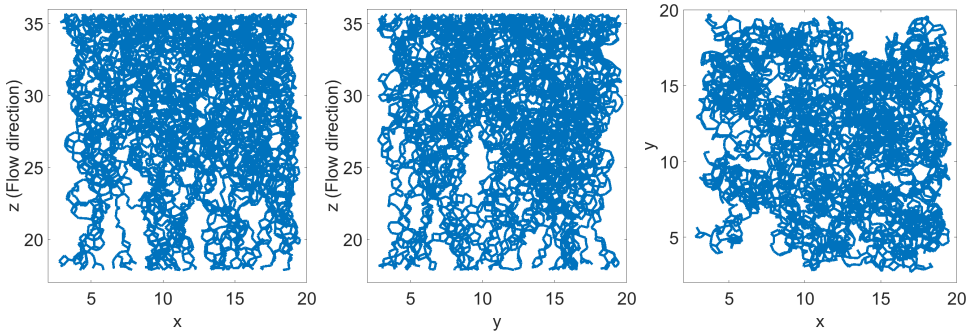
Table 10: Similarity of erodible paths for consecutive simulations.

| $d^n$ | 1&2 | 2&3 | 3&4 | 4&5 | 5&6 | 6&7 | 7&8 | 8&9 | 9&10 |
|---|---|---|---|---|---|---|---|---|---|
| 0.250 | 0.815 | 0.883 | 0.923 | 0.946 | 0.953 | 0.956 | 0.966 | 0.965 | 0.962 |
| 0.255 | 0.709 | 0.847 | 0.908 | 0.920 | 0.920 | 0.909 | 0.935 | 0.937 | 0.947 |
| 0.260 | 0.588 | 0.815 | 0.863 | 0.856 | 0.883 | 0.881 | 0.912 | 0.922 | 0.857 |
| 0.265 | 0.472 | 0.686 | 0.828 | 0.810 | 0.861 | 0.892 | 0.870 | 0.869 | 0.907 |
| 0.270 | 0.309 | 0.741 | 0.646 | 0.756 | 0.850 | 0.892 | 0.885 | 0.931 | 0.904 |
| 0.275 | - | 0.486 | 0.684 | 0.857 | 0.609 | 0.805 | 0.771 | 0.877 | 0.705 |
| 0.280 | - | - | - | - | 0.168 | 0.698 | 0.915 | 0.611 | 0.931 |

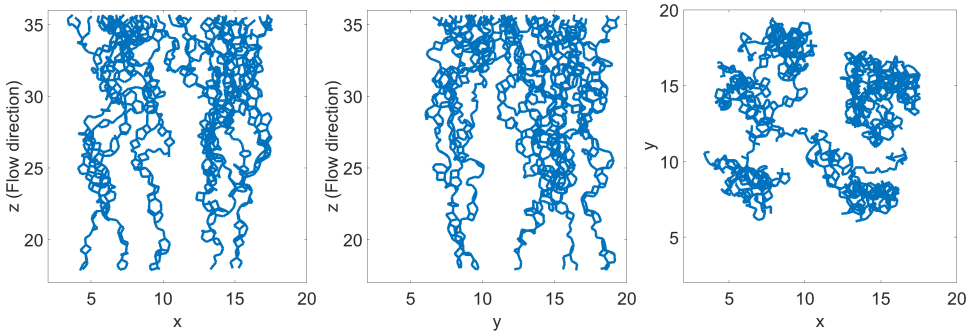
For $d^n \geqslant 0.285$, on the other hand, the filter networks were categorised as seal, i.e. all inlets were clogged at the end of the consecutive simulations. In the previous section, the concept of percolation threshold and directed networks were utilised to assess the values of $d^n_{largest}$; for filter F12dense111 network type 3, $d^n_{largest}$ was predicted to be 0.293. This implies that erodible path(s) exists in the network for $d^n$ up to 0.293, thus allowing some erosion. As discussed, this value agrees well with that obtained through simulations with $d^n_{largest} = 0.29$. Due to premature node clogging, the erodible path(s) was eventually blocked. Finally, the filter was fully effective with no erosion for $d^n \geqslant 0.295$ as expected.

*4.1.4. Effects of Filter Characteristics*

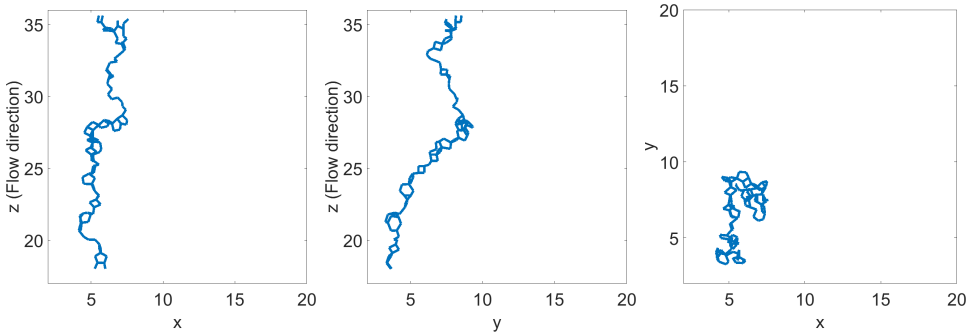
Filter samples of varying $C_u$ and $e$ with size [1,1,1] were used in simulations. In Figure 31(a), $p_{eroded}$ is plotted against $d^n$ for each simulation. In their numerical study of CSDs, Shire & O'Sullivan (2016) showed that the CSDs of random sphere packings with $C_u \leqslant 3$




(a) $d^n = 0.250$.

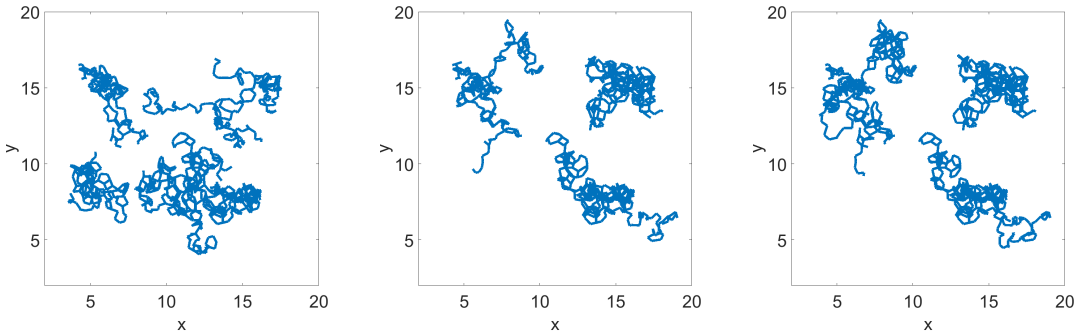



(b) $d^n = 0.265$.




(c) $d^n = 0.280$.

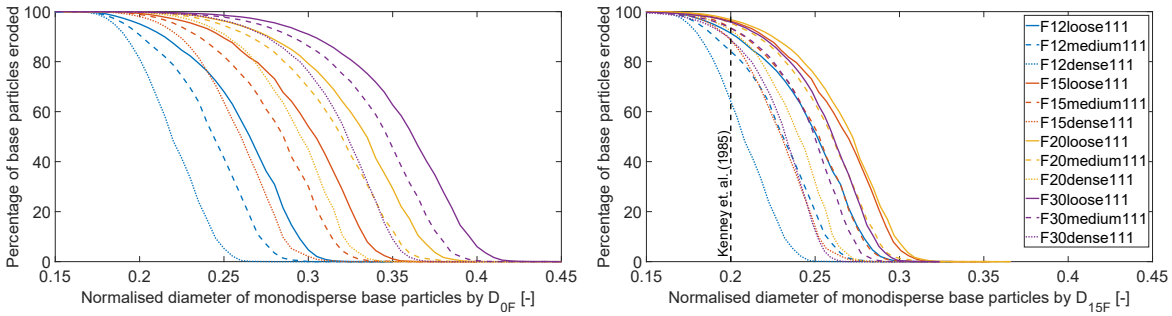
Figure 29: Erodible paths for base particles of different diameters for run 10.




(a) Run 2. (b) Run 6. (c) Run 10.

Figure 30: Plan views of erodible paths for different runs with $d^n = 0.270$.

should plot close to each other when the diameters are normalised by $D_{15F}$ instead of $D_{0F}$. This behaviour is also observed when the curves of $p_{eroded}$ were normalised by $D_{15F}$ as shown in Figure 31(b). Although the values of $d_{largest}$ do not coincide with the experiments by Kenney et al. (1985) i.e. $d_{largest} \approx 0.2D_{15F}$, these findings provide further support for the adoption of $D_{15F}$ as characteristic diameters in filter design criteria. However, it should be noted that the normalised base particle diameters at specific values of $p_{eroded}$ are consistently smaller for dense filters than for loose ones (see Table 11).




(a) Normalised with $D_{0F}$.

(b) Normalised with $D_{15F}$.

Figure 31: Varying filter $C_u$ and $e$ in simulations.

Table 11: Ratio of base particle sizes between dense and loose filters at specific values of $p_{eroded}$.

| $C_u$ | 95% | 80% | 60% | 40% | 20% | 5% |
|---|---|---|---|---|---|---|
| 1.2 | 0.924 | 0.859 | 0.831 | 0.829 | 0.831 | 0.836 |
| 1.5 | 0.915 | 0.879 | 0.858 | 0.854 | 0.856 | 0.857 |
| 2.0 | 0.925 | 0.898 | 0.886 | 0.888 | 0.884 | 0.877 |
| 3.0 | 0.912 | 0.903 | 0.895 | 0.891 | 0.887 | 0.881 |

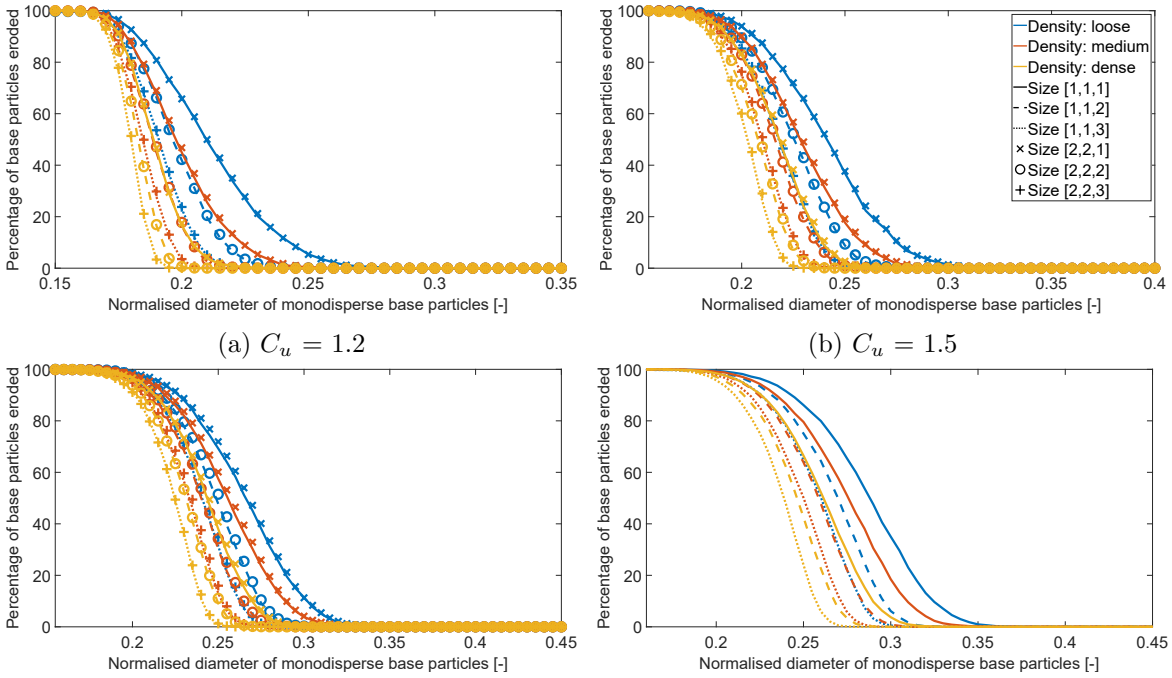
The percolation thresholds, as well as the corresponding $d^n_{largest}$, for these filters were also evaluated in Table 12; the percolation thresholds of these networks have a mean of 0.5788 with relatively small fluctuations, and that the corresponding values of $d^n_{largest}$ agree well with those observed in the simulations. Other behaviours observed in Sections 4.1.2 and 4.1.3 also remained similar. This indicates that the void space topology and the spatial correlations between void space geometry are not affected by filter characteristics.

*4.1.5. Effects of Network Size*

Simulations were performed using networks of varying sizes to investigate how filtration characteristics are influenced by network size. The number of base particles simulated is proportional to the area of the inlet boundary given that base particles were introduced to all inlets for $T_{intro} = 5$. In Section 3.2.2 and Table 12, the percolation threshold, be it based on an undirected or a directed network, is shown to be sensitive to the length-to-width ratio of the network and the use of non-periodic boundaries. However, with reference to Figure 32, it is observed that boundary effects were minimal considering that the results corresponding to networks of the same length, e.g. sizes of [1,1,1] and [2,2,1], are almost identical. This is attributed to the paths taken by the eroded particles being relatively direct, i.e. path tortuousity of around 1.4, such that cases when particles were restricted from moving across boundaries were insignificant.

Table 12: Percolation threshold (and the corresponding $d^n_{largest}$) for each filter network.

| Filter | [1,1,1] | [1,1,2] | [1,1,3] | [2,2,1] | [2,2,2] | [2,2,3] | Mean | Std. |
|---|---|---|---|---|---|---|---|---|
| F12dense | 0.5879 | 0.5968 | 0.6041 | 0.5754 | 0.5953 | 0.5951 | 0.5924 | 0.0098 |
| | (0.293) | (0.288) | (0.285) | (0.299) | (0.290) | (0.290) | (0.291) | (0.005) |
| F12medium | 0.5985 | 0.6166 | 0.6168 | 0.5228 | 0.5677 | 0.5678 | 0.5817 | 0.0363 |
| | (0.341) | (0.332) | (0.332) | (0.377) | (0.356) | (0.356) | (0.349) | (0.017) |
| F12loose | 0.5809 | 0.6103 | 0.6103 | 0.5324 | 0.5495 | 0.5497 | 0.5722 | 0.0334 |
| | (0.379) | (0.364) | (0.364) | (0.403) | (0.395) | (0.395) | (0.383) | (0.017) |
| F15dense | 0.5844 | 0.5998 | 0.6067 | 0.5685 | 0.5923 | 0.5921 | 0.5906 | 0.0132 |
| | (0.343) | (0.336) | (0.333) | (0.351) | (0.340) | (0.340) | (0.341) | (0.006) |
| F15medium | 0.5380 | 0.6111 | 0.6157 | 0.5392 | 0.5581 | 0.6078 | 0.5783 | 0.0372 |
| | (0.421) | (0.381) | (0.378) | (0.421) | (0.411) | (0.383) | (0.399) | (0.021) |
| F15loose | 0.5864 | 0.6148 | 0.6232 | 0.5602 | 0.5710 | 0.5708 | 0.5877 | 0.0258 |
| | (0.422) | (0.408) | (0.402) | (0.437) | (0.432) | (0.432) | (0.422) | (0.014) |
| F20dense | 0.5875 | 0.6064 | 0.6067 | 0.5465 | 0.6037 | 0.6064 | 0.5929 | 0.0239 |
| | (0.381) | (0.370) | (0.370) | (0.404) | (0.373) | (0.372) | (0.378) | (0.013) |
| F20medium | 0.5787 | 0.5802 | 0.5804 | 0.5625 | 0.5815 | 0.5819 | 0.5775 | 0.0075 |
| | (0.431) | (0.430) | (0.430) | (0.440) | (0.431) | (0.430) | (0.432) | (0.004) |
| F20loose | 0.5714 | 0.5773 | 0.5900 | 0.5615 | 0.5753 | 0.5825 | 0.5763 | 0.0097 |
| | (0.457) | (0.454) | (0.448) | (0.464) | (0.457) | (0.453) | (0.456) | (0.005) |
| F30dense | 0.5672 | - | - | - | - | - | 0.5672 | - |
| | (0.429) | (-) | (-) | (-) | (-) | (-) | (0.429) | (-) |
| F30medium | 0.5865 | 0.5898 | 0.5902 | - | - | - | 0.5888 | 0.0020 |
| | (0.469) | (0.467) | (0.467) | (-) | (-) | (-) | (0.468) | (0.001) |
| F30loose | 0.5780 | 0.6054 | 0.6051 | - | - | - | 0.5962 | 0.0157 |
| | (0.501) | (0.484) | (0.484) | (-) | (-) | (-) | (0.490) | (0.010) |
| Mean | 0.5788 | 0.6008 | 0.6045 | 0.5521 | 0.5772 | 0.5838 | 0.5835 | |
| Std. | 0.0152 | 0.0135 | 0.0129 | 0.0177 | 0.0178 | 0.0190 | 0.0093 | |




(a) $C_u = 1.2$ (b) $C_u = 1.5$

(c) $C_u = 2.0$ (d) $C_u = 3.0$

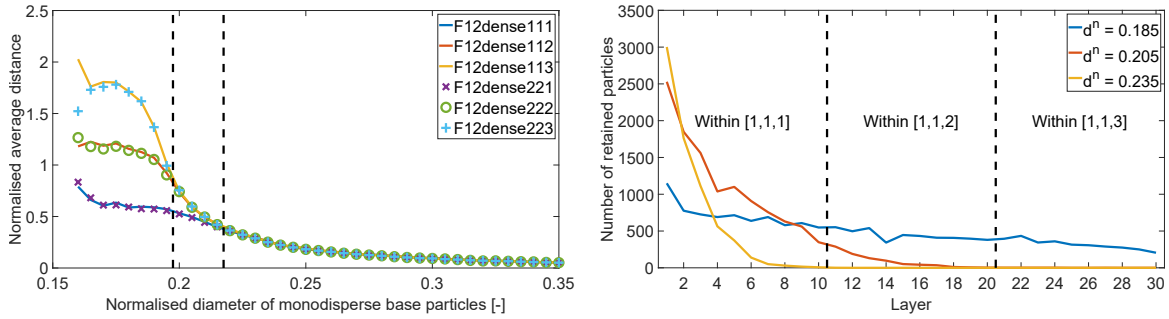
Figure 32: Percentages of base particles eroded with networks of varying sizes.

The increase in network length led to an increase in the percentage of retained base particles as expected, since there were more sites available to capture base particles. However, the hypothesis that $p_{eroded}$ is squared when the filter length is doubled suggested by Roozbahani et al. (2014) is not supported by the results. This shows that the relationship between $p_{eroded}$ and filter length is more complex, e.g. $p_{eroded}$ is also dependent on the number of the particles simulated and the inlets to which the particles were introduced as discussed in Section 4.1.3.

On the other hand, the infiltration depths of the base particles were examined for filters F12dense*. In Figure 33, the average distance travelled by the retained particles normalised by the shortest path in filter F12dense111, as well as the number of particles retained in each layer in the flow direction, are plotted. Three distinct regions are observed: for $d^n \leqslant 0.195$, base particles were retained throughout the depths of the filters. This is because the ratio of the number of base particles simulated to the number of sites available to capture base particles is relatively large such that the networks became saturated very quickly; this ratio decreases with larger $d^n$ as particle straining becomes more dominant. As for $0.2 \leqslant d^n \leqslant 0.215$ and $d^n \geqslant 0.22$, base particles were retained only within sizes of [1,1,2] and [1,1,1] respectively. This agrees with Shire & O'Sullivan (2017), who showed that only the edges close to the inlets were clogged within effective filters.

*4.1.6. Sensitivity Analysis*

Two important assumptions adopted in the network model were studied. Firstly, it is assumed that base particles are influenced only by drag forces exerted by fluid flows, and that base particles can be treated as moving in suspension with fluid. This implies that gravitational force was neglected. Secondly, it is assumed that a node becomes full when the total volume of base particles within the node reaches 50 percent of the node volume. However, the chosen percentage is arbitrary, considering that the void space in each node is irregularly shaped and has a smaller volume-to-surface-area ratio




(a) Normalised distance travelled by particles. (b) Number of retained particles in F12dense113.

Figure 33: Infiltration depths of base particles for filters F12dense*.

when compared to spheres (see Figure 14(c)). Furthermore, it is possible for a retained base particle to prevent another particle from passing before the maximum volume is reached as illustrated in Figure 34, resulting in accelerated node clogging. Therefore, three additional simulations were performed with filter F12dense111, in which (1) base particles were restricted to move only in positive z-direction, (2) each node was considered clogged when a base particle was retained in the node, or (3) both (1) and (2) were applied.

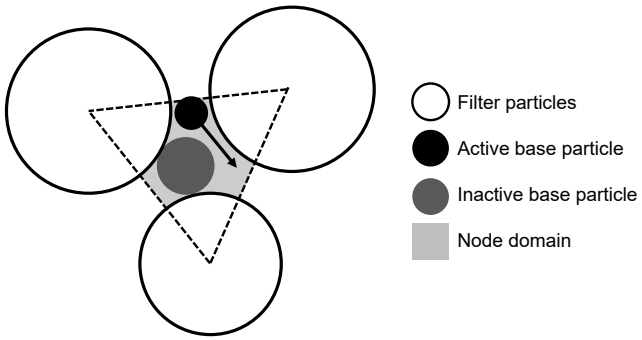
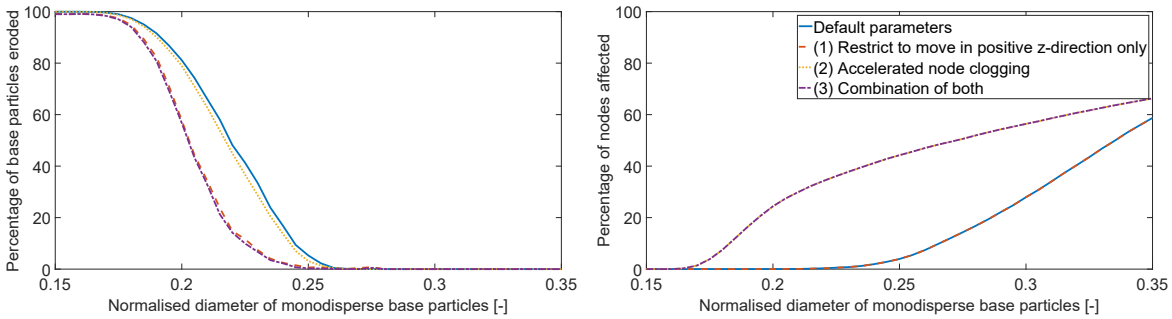



Figure 34: 2D representation of accelerated node clogging.




(a) Percentage of particles eroded. (b) Premature node clogging.

Figure 35: Sensitivity of simulations to additional forces and reduced node volumes.

Figure 35(a) shows that the filter effectiveness is highly sensitive to condition (1), which is equivalent to forcing the base particles to travel in a more direct manner. This resulted in not only an increase in percolation threshold from 0.5879 to 0.6214 (thus the predicted $d^n_{largest}$ decreasing from 0.293 to 0.279), but also a reduction in maximum average path tortuousity from 1.54 to 1.32. The significance of accelerated node clogging, i.e. in

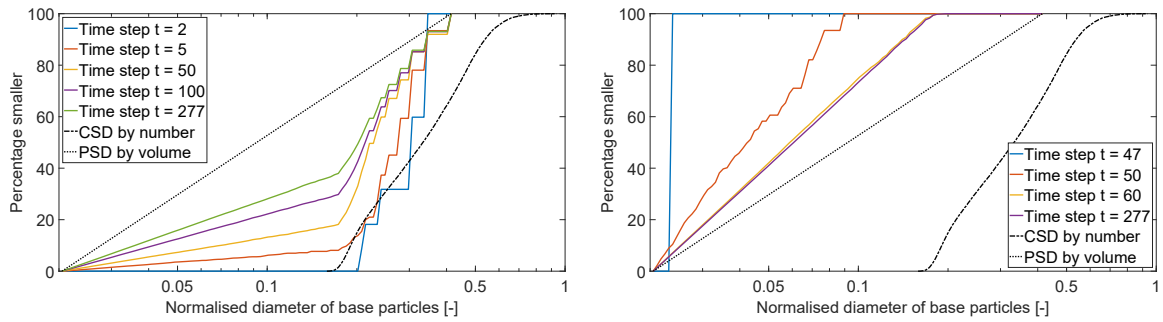
simulations with condition (2), was demonstrated in Figure 35(b) when the percentage of nodes affected by premature node clogging is plotted; a node was considered affected when any of the connected exits was smaller than the simulated base particles. However, the sensitivity of the results to condition (2) is less pronounced compared to (1).

### *4.2. Simulations with Polydisperse Base Particles*

Simulations with polydisperse base particles involved four variables, namely filter $C_u$, filter $e$, base $C_u$ and the ratio of $D_{15F}/d_{85B}$ (as defined in Equation 1). This research considered 504 base-filter combinations using filters of size [1,1,1]; for each filter, the values of base $C_u$ and ratio were varied from 1.5 to 4.5 with an increment of 1.5 and from 0.5 to 7 with an increment of 0.5 respectively. Simulation type 2 was used, i.e. ten consecutive simulations were performed for each filter-base combination.

#### *4.2.1. Effects of Base Coefficient of Uniformity*

The use of polydisperse base particles, i.e. an increase in base $C_u$, allowed the manifestation of self-filtration, which is demonstrated in the evolution of the PSDs of the retained base particles in Figure 36(a). At small time steps, filtration occurred through particle straining such that only the base particles that are larger than the filter constrictions were retained. As more constrictions were clogged, the number of blocked pores increased, resulting in base particles of all diameters to be retained. The PSDs of the retained particles are always below that of the introduced particles since a portion of smaller particles was eroded. Figure 36(b), in which the PSDs of the eroded base particles are plotted, shows that smaller base particles were being eroded initially due to the paths followed being less tortuous. At the end of the simulation i.e. t = 277, the combined PSDs of the retained and eroded particles should be the same as that of the introduced particles.




(a) PSDs of base particles retained.

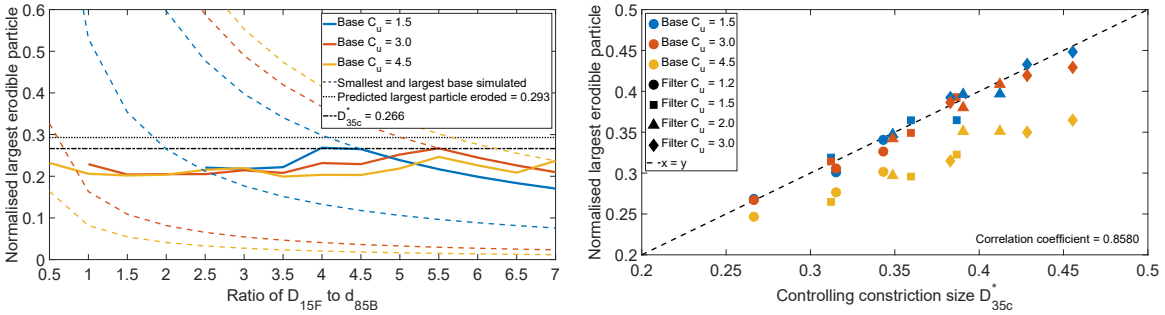
(b) PSDs of base particles eroded.

Figure 36: Evolution of base PSDs for filter F12dense111, base $C_u = 4.5$ and $D_{15F}/d_{85B} = 4$.

In Figure 37(a), the largest erodible base particles for simulations using filter F12dense-111 are shown to be consistently smaller than that predicted based on percolation threshold. Unlike in monodisperse simulations, the number of base particles within a specific size range is reduced. Therefore, the probability of the largest theoretical base particles following the erodible paths is significantly lowered. Furthermore, given that base particles larger than the theoretical value were also introduced, it is easier for the nodes to become full prematurely.

It is interesting to note that the values of $D^*_{35c}$ constitute the upper limits for the diameters of the largest erodible particles for all simulations as demonstrated in Figure 37(b). This suggests that the concept of percolation threshold can provide an alternative explanation for the adoption of the 35th percentile (100 minus percolation threshold) of

the CSD as the controlling constriction size in Equation 4. It should be noted that the use of $D^*_{35c}$ in the criterion is originally based on the probability of a particle moving forward in a cubic network derived analytically by Locke et al. (2001). However, cubic networks are not able to describe accurately the void space structure of granular filters as discussed in Section 4.1.2.




(a) $d^n_{largest}$ for filter F12dense111.

(b) Controlling constriction size of $D^*_{35c}$.

Figure 37: Comparison of largest erodible particles against predicted values.

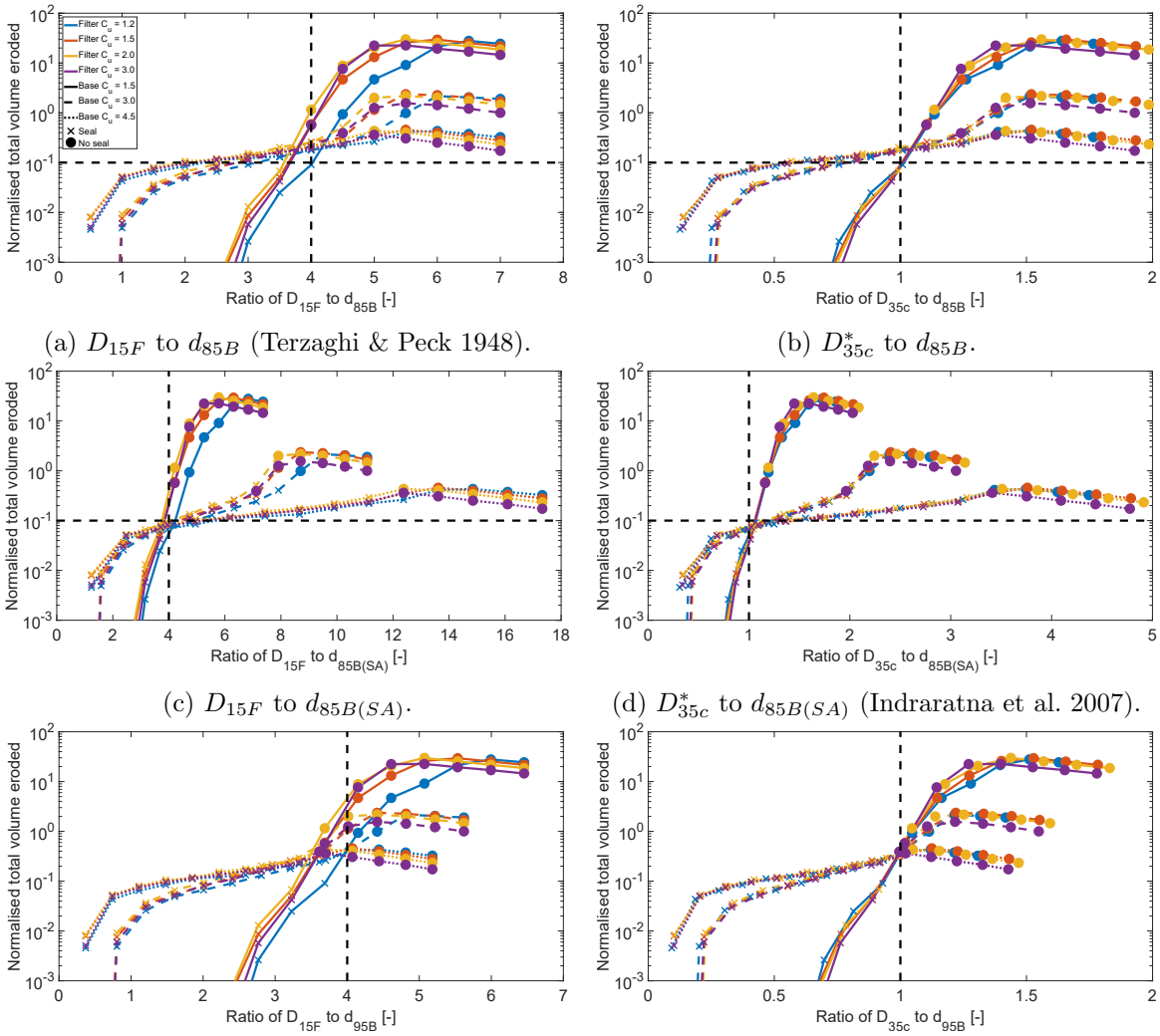
*4.2.2. Assessment of Granular Filter Design Criteria*

To assess the accuracy of the granular filter design criteria set out in Equations 1 to 4, the proposed ratios were applied to the results. Figures 38 and 39 present the results for dense filters with varying $C_u$, and for $C_u = 1.2$ filters with varying $e$ respectively; Figure 40 presents the results for all simulations. The total volume eroded was normalised by the cross-sectional area of each filter, and the plateaus in total volume eroded at larger ratios were caused by a limit on the number of base particles that were simulated simultaneously at 500,000 to reduce computational time. The framework proposed by Foster & Fell (2001) is once again adopted as shown in Table 13; boundaries (1) and (2) refer to those between categories (1) and (2), and (2) and (3) respectively.

Table 13: Characterisation of filter effectiveness in polydisperse simulations (Foster & Fell 2001).

| Category | Filter Effectiveness | Total Volume Eroded | Fully-Filled Inlets |
|---|---|---|---|
| 1 | Seal with no erosion | Less than 0.1 | All |
| 2 | Seal with some erosion | More than 0.1 | All |
| 3 | No or partial seal with continuous erosion | More than 0.1 | No or some |

Previous studies revealed that filters that were designed to Equation 1 had varying effectiveness depending on base $C_u$. For uniform base particles, filters with $D_{15F}$ to $d_{85B}$ ratios higher than 4 can be effective (Sherard et al. 1984). On the contrary, filters with lower ratios had excessive soil loss in the cases of well-graded base particles (Lafleur 1984). This was also observed in the simulations as shown in (a) and (b), specifically those plotted in the second quadrant. Indraratna et al. (2007) therefore suggested the use of $d_{85B(SA)}$ for boundary (1) (see (c) and (d)) to ensure the presence of sufficient number of larger base particles. This is because when the base PSD by volume is considered, the contribution of the larger particles to filtration is misrepresented (Indraratna et al. 2007). As for boundary (2), the adoption of $d_{95B}$ is more accurate since it is the largest base




(a) $D_{15F}$ to $d_{85B}$ (Terzaghi & Peck 1948).

(b) $D^*_{35c}$ to $d_{85B}$.

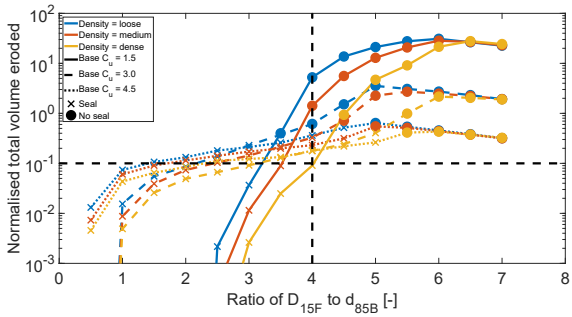
(c) $D_{15F}$ to $d_{85B(SA)}$.

(d) $D^*_{35c}$ to $d_{85B(SA)}$ (Indraratna et al. 2007).

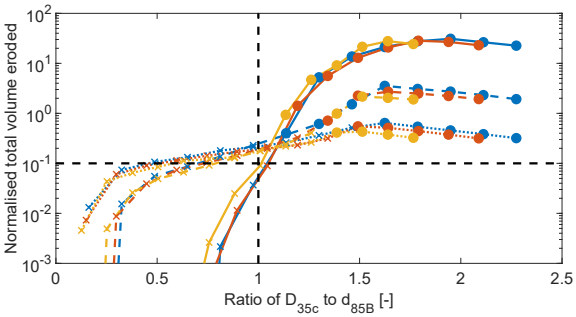
(e) $D_{15F}$ to $d_{95B}$ (Foster & Fell 2001).

(f) $D^*_{35c}$ to $d_{95B}$.

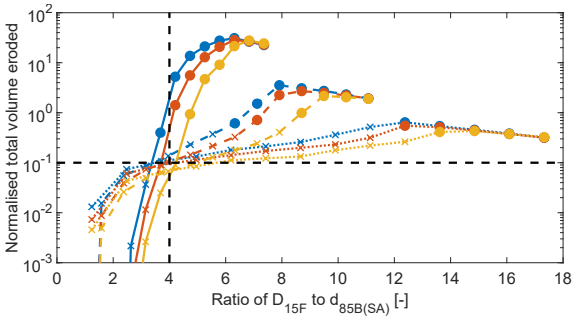
Figure 38: Application of granular filter design criteria to simulations with dense filters of varying $C_u$.



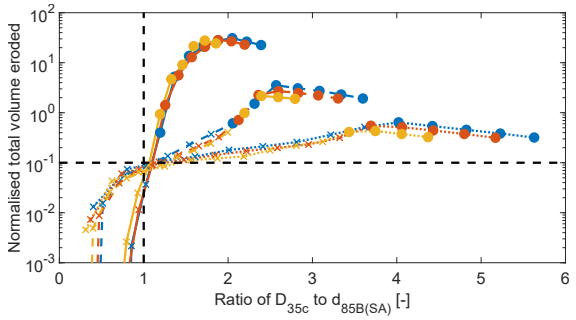

(a) $D_{15F}$ to $d_{85B}$ (Terzaghi & Peck 1948).



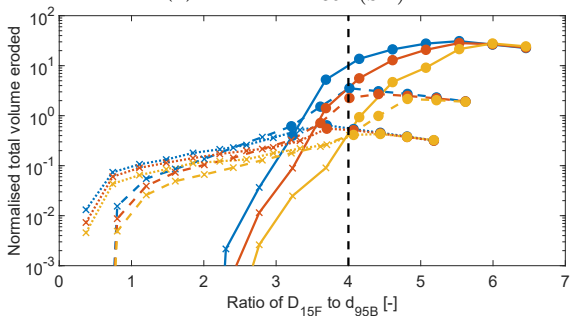

(b) $D^*_{35c}$ to $d_{85B}$.



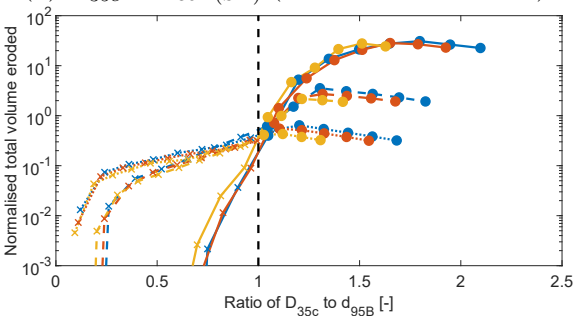

(c) $D_{15F}$ to $d_{85B(SA)}$.




(d) $D^*_{35c}$ to $d_{85B(SA)}$ (Indraratna et al. 2007).

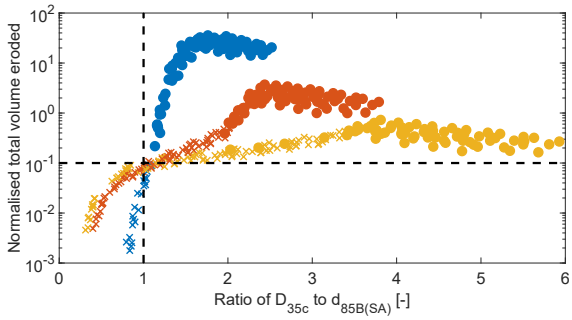



(e) $D_{15F}$ to $d_{95B}$ (Foster & Fell 2001).

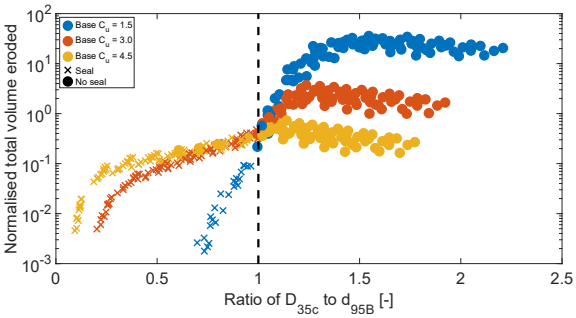



(f) $D^*_{35c}$ to $d_{95B}$.

Figure 39: Application of granular filter design criteria to simulations with constant filter $C_u = 1.2$ and varying filter $e$.




(a) $D^*_{35c}$ to $d_{85B(SA)}$ (Indraratna et al. 2007).




(b) $D^*_{35c}$ to $d_{95B}$.

Figure 40: Application of granular filter design criteria to all 504 simulations.

particles that control the eventual sealing of filters as shown by Shire & O'Sullivan (2017) (see (e) and (f)). However, these results agree only qualitatively with Equations 2 and 3.

The use of $D_{15F}$ in (a), (c) and (e) shows that there is a limited unifying effect, i.e. the curves do not converge completely for varying filter $C_u$ and $e$ as discussed in Section 4.1.4. On the other hand, $D^*_{35c}$ accounts for the effects of filter characteristics directly, and thus provides higher explanatory power as demonstrated in (b), (d) and (f). This is further supported by the void space topology and the spatial correlations being similar for all filters as demonstrated in Section 4.1.4. However, it should be noted that Indraratna et al. (2007) derived filter CSDs analytically using the inscribed circle method (ICM), whereas the research adopted the DEM-DTM method. Therefore, the values of $D^*_{35c}$ are inconsistent as shown in Figure 41.

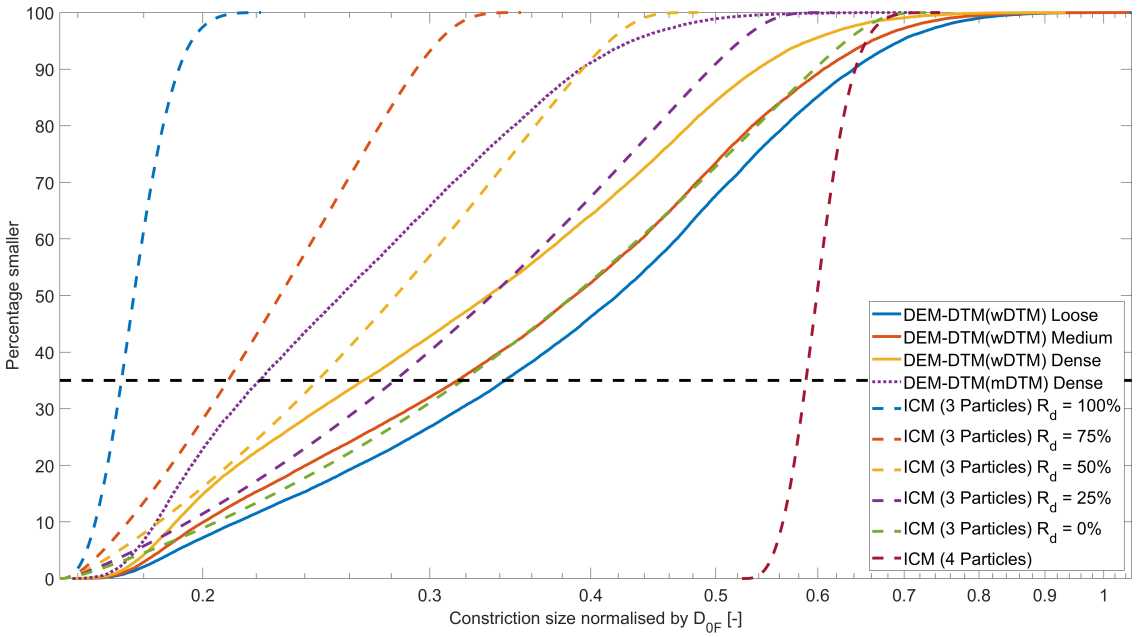



Figure 41: Comparison of CSDs obtained using the inscribed circle method (ICM) and the DEM-DTM Method for filter with $C_u = 1.2$.

## 5. Conclusion

This research has proposed a network model developed based on DTM-derived data, which allowed the implementation of two novel features: (i) realistic fluid flows, and (ii) accurate void space structure. The model was coupled with a flow-biased walk algorithm to simulate the movement of base particles within the filter void space. In regard to the simulated diameters of the largest erodible particles, the proposed network model showed very good quantitative agreement with geometrical DEM simulations by Roozbahani et al. (2014), but only qualitative agreement with the experimental filter tests by Kenney et al. (1985).

It is demonstrated that the effectiveness of filters in retaining base particles can be fully explained when the concepts of percolation threshold and premature node clogging were considered in conjunction. Both of these properties are directly influenced by the characteristics of the adopted network, highlighting the importance of accurate description of void space structure. Therefore, applying idealising assumptions that oversimplify void space structure, i.e. the use of regular lattice and randomised geometry, are not physically justifiable. It is interesting to note that networks constructed from both the wDTM and the mDTM produced comparable results. However, the wDTM is favoured over the mDTM given the lower computational demand.

The simulation results provided insights into the fundamental mechanics of filtration process, as described below:

1. The existence of spatial correlation between geometric properties in random sphere packings increases the probability to encounter clusters of smaller pores with smaller connected constrictions that are more susceptible to clogging.

2. The tortuousity of erodible paths increases with both base particle diameter $d$ and time.

3. Larger base particles follow a smaller number of common erodible paths in line with Roozbahani et al. (2014).

4. The erodible paths followed by monodisperse base particles in ineffective filters remain similar, especially in the long term.

5. The percentage of particles eroded $p_{eroded}$ is not squared when the filter length is doubled as suggested by Roozbahani et al. (2014); $p_{eroded}$ is also dependent on the number of base particles and the inlets by which the base particles enter a filter.

6. The infiltration depth decreases with increasing $d$; base particles are retained very close to the inlets in effective filters, which is consistent with Shire & O'Sullivan (2017).

The research considered 504 filter-base combinations in network model simulations to assess the accuracy of granular filter design criteria; the filter effectiveness classification framework proposed by Foster & Fell (2001) was adopted. Similar to experiments by Lafleur (1984), excessive soil loss was observed in simulations for base particles with large $C_u$ when filters were designed based on the conventional $d_{85B}$. This is because the contribution of the larger particles to filtration is misrepresented when PSD by volume is considered (Indraratna et al. 2007). The use of $d_{85B(SA)}$, which is smaller than $d_{85B}$, suggested by Indraratna et al. (2007) was found to best divide (1) seal with no erosion and (2) seal with some erosion. On the other hand, the use of $d_{95B}$ proposed by Foster

& Fell (2001) formed a better boundary between (2) seal with some erosion and (3) no seal with continuous erosion, since it is the largest base particles that govern the eventual sealing of filters.

The use of $D_{15F}$ resulted in a limited unifying effect for results with varying filter $C_u$ and $e$. Constriction based method, i.e. $D^*_{35c}$, proposed by Indraratna et al. (2007) accounts for the effects of filter characteristics directly, and thus provided higher explanatory power. It was shown that the void space topology and the spatial correlation between geometric properties within granular filters are not affected by filter characteristics, such that the percolation thresholds remain similar at 58% (which is increased to 65% to account for premature node clogging). Furthermore, the values of $D^*_{35c}$ constitute the upper limits for the diameters of the largest erodible particles for all simulations. This suggests that the concept of percolation threshold can provide an alternative explanation, compared to Locke et al. (2001), for the adoption of the 35th percentile (100 minus percolation threshold) of the CSD as the controlling constriction size. However, the inconsistency in the methods adopted to obtain filter CSD, and thus $D^*_{35c}$, should be noted when comparing with Indraratna et al. (2007).

The proposed model consistently allowed larger base particles to be eroded compared to that observed in experiments by Kenney et al. (1985). The research showed that the filter effectiveness is sensitive to external forces such as gravity. Other factors include the filter CSD being affected by the assumption of spherical filter and base particles as shown by Taylor (2016), and the simulations considering only the extreme case in which base particles were introduced to all inlets. These factors can be further examined in future research. On the other hand, two possible applications of the network model include investigating how filter anisotropy affects filtration characteristics by considering sheared filter samples, and determining how filters can be optimised by varying its particle size with depth.

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## Appendix A - Validations

*Constriction Sizes*

The algorithm responsible for the calculation of constriction sizes was validated by comparing the distributions with those obtained by Shire (2014) as shown in Figure 42.

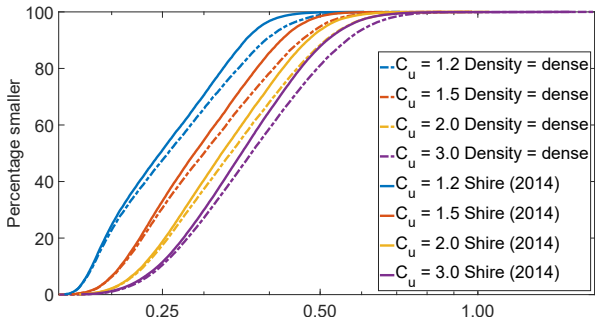



Figure 42: Comparison of edge diameter (mDTM) against those obtained by Shire (2014).

*Flow Solver*

Validations were performed to check the accuracy of fluid flows calculated. Firstly, the values of the local pressures in the nodes $[P]$ were compared with those obtained with an open-source pore network modelling package written in Python named OpenPNM using Hagen-Poiseuille conductance model (Gostick et al. 2016). For filter 12dense111(wDTM), the maximum percentage difference was found to be $2.8 \times 10^{-4}\%$.

Secondly, regular simple cubic packings of monodisperse particles with varying $\phi$ were generated. The values of the permeability k defined in Equation 13 were compared against those given in other researches as shown in Figure 43. Martin et al. (1951) obtained the value of k through experimental studies of pressure drop through regular packings. Snyder & Stewart (1966), Sørensen & Stewart (1974) and Zick & Homsy (1982) on the other hand, calculated k using numerical methods. The values of k can also be obtained using Kozeny-Carman estimate (Sufian et al. 2019):

$$k = \frac{(1-\phi)^3}{5\phi^2 S_v^2}, S_v = \frac{\sum_i \pi d_i^2}{\sum_i \frac{\pi}{6} d_i^3} \tag{15}$$

where $S_v$ is the specific internal surface area to volume ratio obtained from the complete PSD and $d$ is the particle diameter. For the densest simple cubic packings of which $\phi = 0.5236$, the permeability k calculated with the adopted conductance model is in good agreement with those obtained through other approaches. Although the accuracy of the adopted model decreases as $\phi$ reduces, the predicted values are within 20% of Zick & Homsy (1982) down to $\phi = 0.483$.

Thirdly, the permeabilities k of random packings, specifically the virtual filter samples with size [1,1,1], were compared against those calculated using Kozeny-Carman estimate (Equation 15). As illustrated in Figure 44, similar trend, i.e. the accuracy increasing with the filter density, is observed. All values calculated with the adopted conductance model are within 25% of Kozeny-Carman estimate.

These validations proved that the algorithm to calculate fluid flows was implemented correctly. Furthermore, they showed that the adopted local conductance model (Equation 10) was sufficiently accurate for the purpose of simulating filtration processes, in which the movements of particles were influenced by the magnitude and direction of fluid flows.

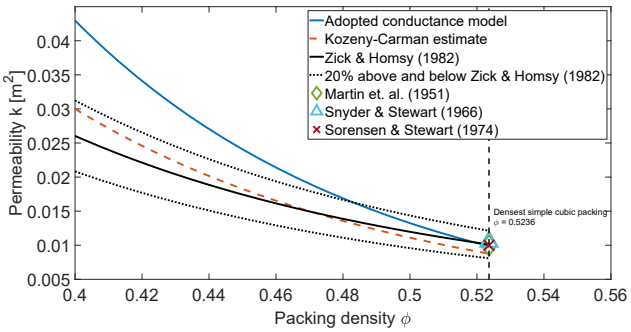
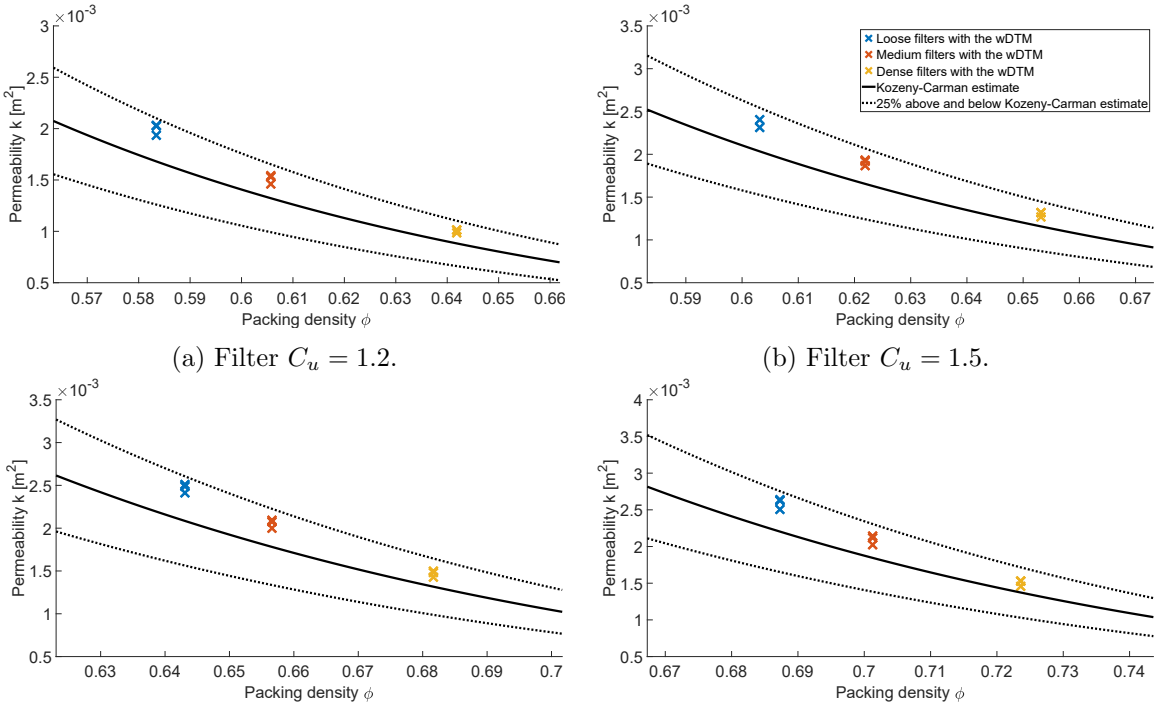



Figure 43: Permeabilities in regular simple cubic packings.




(a) Filter $C_u = 1.2$.

(b) Filter $C_u = 1.5$.

(c) Filter $C_u = 2.0$.

(d) Filter $C_u = 3.0$.

Figure 44: Permeabilities in random packings.

However, the modifications applied to accommodate for the presence of base particles were not validated.

*Overall Simulation*

With reference to Shire (2014), the scripts implemented responsible for the introduction of base particles and the simulation of particle movements were validated by applying the flow-biased walk algorithm with monodisperse base particles having diameters equal to constriction sizes $D^*_{3c}$ and $D^*_{5c}$. As shown in Table 14, the corresponding percentages of edges that were clogged at the end of simulations should be approximately 3% and 5% respectively. Small percentages were chosen so that the base particles were able to move through the whole depth of the filter. These simulations were executed using filter F12dense111, and were repeated five times to ensure the results were repeatable.

Table 14: Percentage of edges that were clogged at the end of simulations.

| Base Particle Diameter | Run 1 | Run 2 | Run 3 | Run 4 | Run 5 | Standard Deviation |
|---|---|---|---|---|---|---|
| $D^*_{3c}$ | 2.9627 | 2.9602 | 2.9602 | 2.9627 | 2.9602 | 0.0014 |
| $D^*_{5c}$ | 4.8917 | 4.8892 | 4.8892 | 4.8917 | 4.8917 | 0.7881 |

*Repeatability of Simulations with Monodisperse Base Particles*

Given the ability of the network model to simulate several stochastic behaviours of the base particles, multiple simulations might be required to produce meaningful statistics. However, the results obtained with the adopted network model were shown to be replicable when the same simulations for filter F12dense111 were repeated ten times. Figure 45 shows that the randomness in the sequence by which base particles were introduced and in the path followed based on flow-biased probabilities did not produce significant variations.

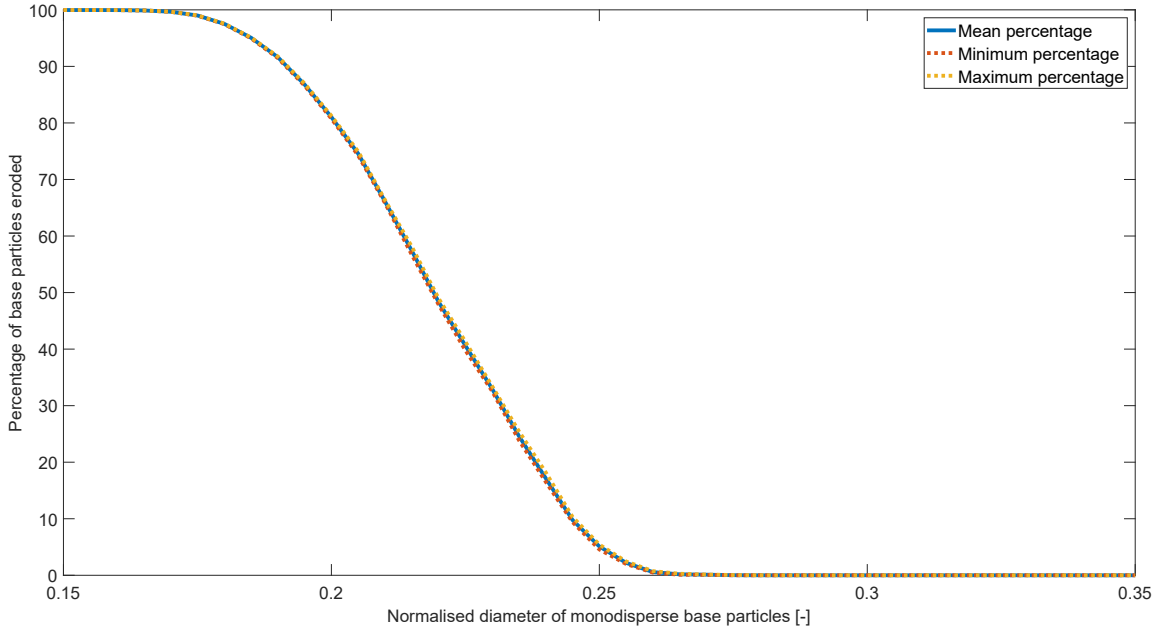



Figure 45: Repeatability of monodisperse simulations using F12dense111.